

Summary - Bioinformatics

Introduction to Molecular biology	7
DNA molecule (in Human beings)	7
DNA - bases	7
Molecular Biology	8
Central Dogma of Molecular Biology	8
Transcription & Translation	9
RNA Polymerases	10
RNA Major Classes	10
Ribosomes	11
Protein basic units: the ammino acids	11
Gene Structure	12
DNA Chromatin Structure	13
Alternative Splicing	14
Sequencing Splicing	15
Splicing Effects on Alignment	16
Mutations	16
Gene fusion mechanisms	17
Computational Issues	18
Proteins & effects of mRNA aberrations on proteins	18
Impact of point mutations on protein's chain	18
Protein folding: Levels of Protein Structure	19
Protein's Domains	19
Bioinformatics	20
Alignment algorithms	20
Alignment Tools	20
Relationship alignment-biology	21
Sequence similarity	21
BLAST - Basic Local Alignment Search Tool	22
BLAST - seed and extend	22
Speed of BLAST	23
NEXT GENERATION SEQUENCING: Format and Tools	23
Data Flow	24
Data Formats	24
Fasta Format	24
MultiFasta Format	25
Fastq Format	25

SAM Format	26
Genome Annotation Format	28
GFF Format	28
BED Format	29
Tools For File Elaboration	30
SAMTools.....	30
BEDTools	30
Alignment Tools.....	31
Sequence Alignment.....	31
Bowtie.....	31
BLAST.....	32
TopHat – splicing aware alignment tool	32
Analysis and Visualisation Tools	33
SeqMonk	33
NGS TECHNOLOGY	35
DENATURATION/HYBRIDIZATION	35
NUCLEIC ACID THERMODYNAMICS SINGLE 2 DUPLEX	35
POLYMERASE CHAIN REACTION (PCR).....	35
DNA LIBRARY PREPARATION.....	36
Data and Statistics	38
Statistics.....	38
Variability.....	39
Sampling.....	39
Data exploration.....	39
Variables	39
Charts	40
Techniques	43
Biological network.....	46
BASICS on Bio	47
Biological Network Model.....	48
Types of Networks	50
Interaction Network	50
Pathway Networks	51
Regulatory Network	52
Expression Network	52
Biological Network Property.....	52
Protein Interaction Network.....	53
Network Databases	54

Pathway Databases	55
KEGG	55
Applications	56
Network Reconstruction	56
Markov Chain Clustering (MCL)	57
DATA INTEGRATION.....	58
Statistical-Algorithmic Method for Biclust Analysis.....	58
NETWORK-BASED CLASSIFICATION	59
Network Alignment	60
Recap	61
PPI Networks Properties	61
Biological Network.....	61
Protein-Protein Interaction (PPI)	62
Structural Properties	62
Network Weighting Schemes	63
GENE ONTOLOGY - GO	63
Semantic Similarity	63
Functional Module Identification	63
Hub Protein Identification	64
Lethality	64
The Dijkstra's algorithm	65
NETWORK IN PRACTICE some real example.....	66
Which microRNAs to choose for a more targeted experiment?	66
Pathways interaction	67
Genes miRNAs interactions	69
Overview of Machine Learning and Pattern Recognition	73
Introduction	73
Data provided by sequencing	73
Machine learning and sequencing.....	73
Artificial Intelligence.....	73
Machine Learning.....	74
Pattern Recognition	74
What is Machine Learning?.....	74
Knowledge-based vs. Learning Systems.....	74
Why "Learn"?.....	74
Types of machine learning	75
Supervised Learning	75
Reinforcement learning	76

Unsupervised Learning	77
Classification problems: main concepts	78
Feature vector	78
Feature space.....	79
The concept of class	79
Representing data in a ML system.....	79
Representation CAVEAT	79
Building a ML system	80
Misclassification error	80
Learning curves.....	80
Training/Test set partitions	81
Crossvalidation	81
Internal cross validation	81
Naïve Rule.....	82
Confusion matrix for binary classification.....	82
Quantifying performance	82
Error rate – binary classification	82
Precision and Recall	82
Classification parameters.....	83
Principal Component Analysis (PCA)	84
Linear Discriminant Analysis (LDA).....	84
Feature selection vs Dimensionality reduction	85
Steps.....	85
Evaluation	85
Naïve Search.....	88
Sequential forward selection (SFS) (heuristic search)	88
Sequential backward selection (SBS) (heuristic search)	88
SFS vs SBS.....	89
Bidirectional search (BDS).....	90
“Plus-L, minus-R” selection (LRS).....	90
Genetic Algorithms (GAs).....	90
Explore new solutions.	91
Genetic Algorithms	91
Feature Selection using GAs.....	92
Summary of ML problems.....	92
The classification framework	93
Choosing a classifier	93
Overfitting/Underfitting.....	93

Bias-Variance Trade-off.....	94
Generative vs. Discriminative Classifiers	94
Classification	95
Voronoi diagrams	96
Bayes' Theorem: Basics.....	96
Prediction Based on Bayes' Theorem.....	96
Classification: deriving max Posteriori	97
Avoiding the Zero-Probability Problem	98
Linear classifier	99
Nonlinear SVMs	100
What about multi-class SVMs?.....	102
SVMs: Pros and cons.....	102
Decision trees	103
Induction of Decision Trees.....	103
TDIDT Algorithm	104
Attribute Selection Criteria	105
Information-Theoretic Approach	105
Entropy	105
Residual Information.....	106
Limitation	107
Information Gain Ratio	107
Gini Index	107
Three measures of impurity	108
Decision trees	108
Ensemble classifiers	108
General Idea	108
Main challenge: promote diversity	108
Ensemble methods	109
Neural Networks	112
NN - Learning.....	112
NN - architectures	112
The neuron	112
Feed-forward single layer NN	113
Perceptron: learning rule	113
Gradient descent	114
Feed-forward multi layer	114
Backpropagation	114
Limitations of Neural Networks.....	118

Deep learning..... 118

 Learning Hierarchical Representations 119

Convolutional Neural Network (CNN)..... 120

 CNN Architecture 120

 Multiple Convolutions 121

 Convolutional Layer 121

 Non-linear Layer 122

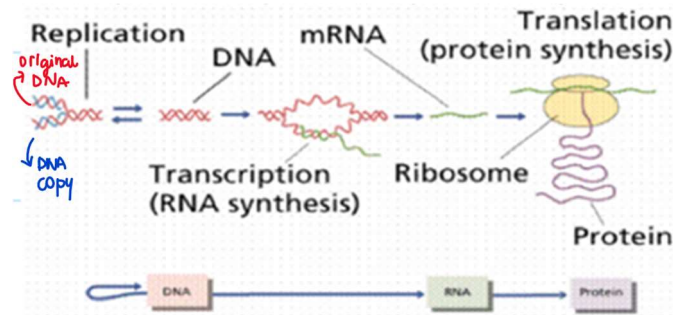
 Pooling Layer 122

Molecular Biology

DNA: DNA → DNA (Replication)

RNA: DNA → RNA (Transcription / Gene Expression)

Protein: RNA → Protein (Translation)



Replication -> **DNA synthesis**, melting, PARENT1 + PARENT2 =newborn,

Transcription -> **RNA synthesis**, most common operation because all need functionality,

Translation -> **production of proteins**, 95% of the time involved in this operation.

Note: duplication of cells is the only occasion that we have DNA duplication, the new cell is a clone.

Central Dogma of Molecular Biology

“Genetic information flows from DNA to RNA to proteins”

DNA synthesis (replication) -> Cell duplication

DNA transcription (activation) -> Gene Expression

Gene Expression

There are 4 major events that occur during the process of gene expression:

- Transcription,
- RNA processing,
- Translation,
- Protein processing.

DNA → Transcription → RNA → Translation → PROTEINS

Terms

DNA can be represented as a sequence of letters (A, C, G, T).

Different levels of subpart of the DNA:

- **Base pair (bp)** - one pair of DNA bases (1 letter),
- **Gene** - section of DNA that produces a functional product,
- **Chromosome** - physical linear sequence of DNA, physical organization of the DNA,
- **Genome** - entire collection of DNA for an organism

E Coli	1 chromosome	5 x 10 ⁶ bases (5 Mbps)
Drosophila	8 chromosomes	2 x 10 ⁸ bases (200 Mbps)
Human	48 chromosomes	3 x 10 ⁹ bases (3 Gbps)

- human → 46

- The bacteria are composed by one cell only, 1 circular genome, less complexity overall.

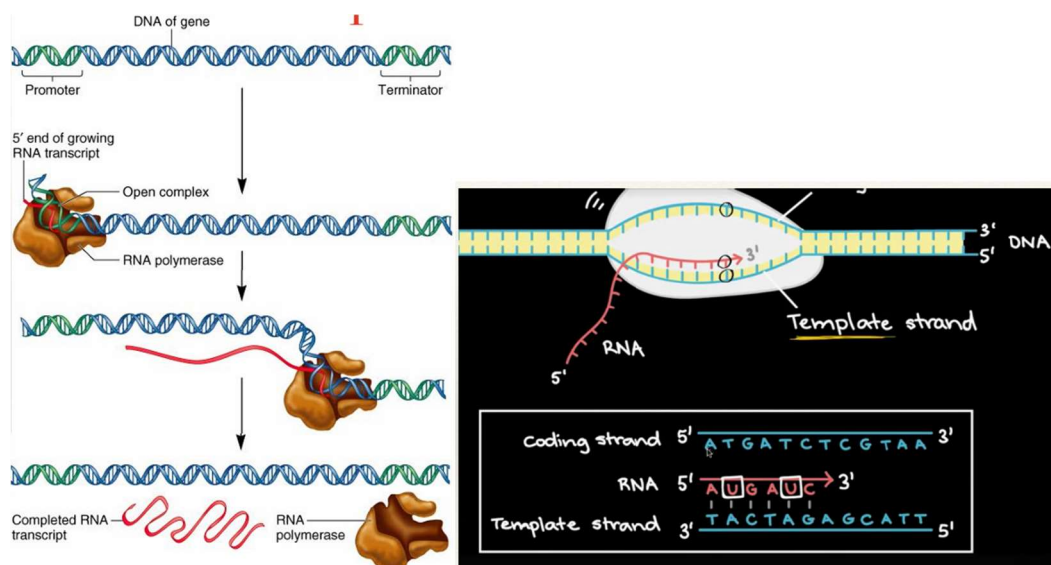
Transcription & Translation

DNA synthesis (replication) maintains the genetic information and passes this to the next generation. **RNA synthesis (transcription)** is a transfer of the information from the DNA where it is stored into RNA which can be transported and interpreted. The language (of nucleic acids) is essentially the same. Messenger RNA (mRNA) moves the information from the DNA to the ribosomes to enable the **production of protein (translation)**. Translation represents a change in the language from the nucleotide letters in the RNA to the amino acid letters in the protein.

Note: not all the pieces of the DNA become a mRNA.

Transcription DNA > RNA

1. **Initiation:** RNA polymerase moves along the DNA in synthesizing a RNA transcript. Synthesis is 5' - >3'. Only 1 strand of DNA is read as a template,
 - RNA polymerase binds to a specific region of DNA, called the **promoter**,
 - In prokaryotes, RNA polymerase directly recognizes the promoter,
 - In eukaryotes, **transcription factors** are needed to help RNA polymerase bind,
 - Once bound, the DNA double helix disaggregates locally to form a **transcription bubble**,
 - The polymerase begins to synthesize RNA from the DNA template strand.
2. **Elongation:** Transcription factors and RNA polymerase recognize and bind the promoter DNA adjacent to the **promoter** (promoter = regulatory element) is denatured forming the open promoter complex,
 - RNA polymerase advances along the DNA, synthesizing the new RNA strand in the 5' → 3' direction,
 - It adds complementary nucleotides to the DNA template sequence,
 - The DNA behind the enzyme (RNA polymerase) rewinds as the RNA is lengthened,
3. **Termination:** A termination signal is reached causing RNA polymerase to dissociate from the DNA,
 - Transcription stops when RNA polymerase encounters a termination sequence.
 - In prokaryotes, this can happen in two ways:
 - Protein-dependent (e.g. Rho factor),
 - Protein-independent, via a hairpin loop followed by a series of uracils,
 - In eukaryotes, the process is more complex and involves termination signals and post-transcriptional modifications (such as polyadenylation).



➔ It reads the blue strand and produces the red strand.

RNA Polymerases

- Eukaryotes:
 - 3 RNA Polymerases I, II, and III involved in eukaryotic transcription – each has ~12 subunits
 - **RNA polymerase I** (localized to the nucleolus) transcribes the rRNA precursor molecules,
 - **RNA polymerase II** produces most mRNAs and snRNAs,
 - **RNA polymerase III** is responsible for the production of pre-tRNAs, and other small RNAs,
- ➔ **3 sets of basal transcription factors** recognize promoter DNA sequences.

Basal transcription factors are regulatory proteins that:

- Recognize specific nucleotide motifs in the promoter (e.g., the TATA box),
- Correctly position RNA polymerase II on the DNA,
- Open the double helix and initiate assembly of the transcription complex.

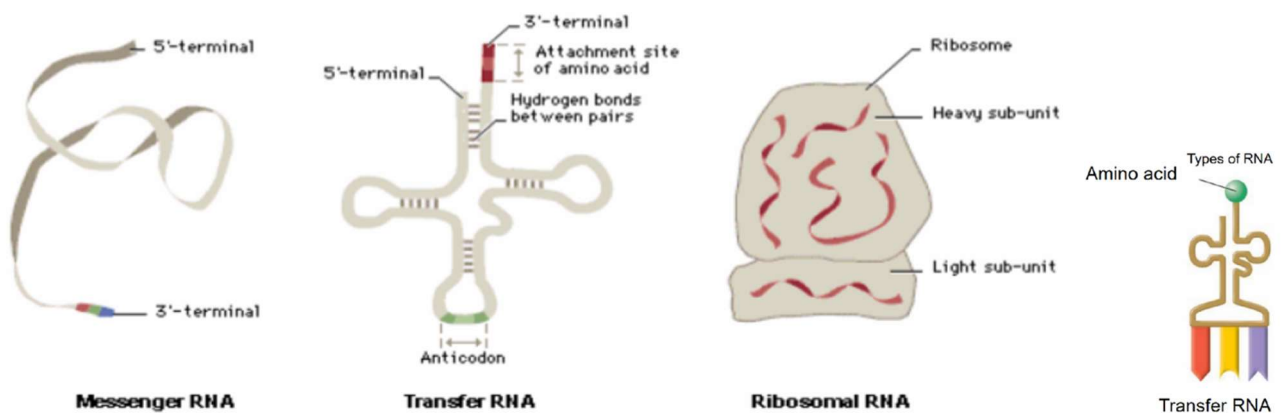
RNA Major Classes

3 major classes of RNA (mRNA, tRNA, & rRNA) synthesized by transcription of the appropriate genes and are involved in protein synthesis. All three major types of RNA are involved in directing the formation of protein.

- **Messenger RNA (mRNA):** carries the message from the DNA to the ribosome.
- **Ribosomal RNA (rRNA):** are major structural components of the protein-synthesizing ribosome, fully inherited from the mother, makes up the ribosomes that reads the mRNA to build the correct amino acid sequence
- **Transfer RNA (tRNA):** act as adaptor molecules in aligning the amino acids according to the sequence present in the mRNA,
 - brings the amino acids to the ribosome; during protein construction, transfer RNA (tRNA) transfers each amino acid to the ribosome.

Non coding RNA – functional RNA molecules that are not translated into protein, e.g. ribosomal RNA (rRNA), transfer RNA (tRNA), microRNA (microRNAs are predicted to control the translational activity of approximately 30% of all protein-coding genes in mammals).

- ➔ Long non-coding RNA,
- ➔ Short non-coding RNA.

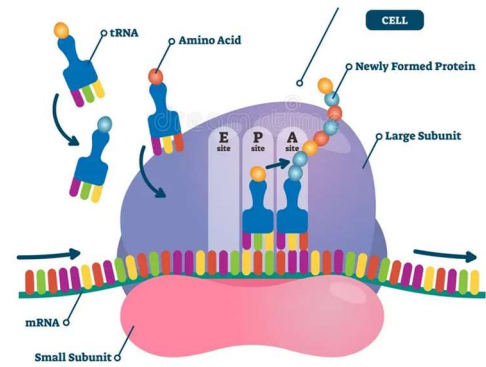
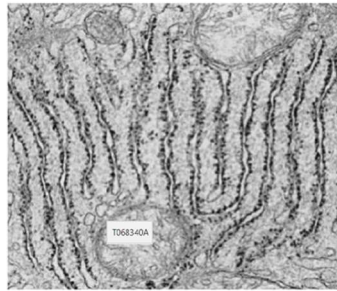


Ribosomes

Ribosome



Ribosomal RNA



Ribosomes are made up of proteins and ribosomal RNA (rRNA).

- **Site A – Aminoacyl-tRNA site:** this is where a new charged tRNA (aminoacyl-tRNA) arrives with its specific amino acid, complementary to the mRNA codon,
- **P Site – Peptidyl-tRNA site:** contains the tRNA that carries the growing polypeptide chain. It is the catalytic center where the peptide bond is formed between the new amino acid (in A) and the existing chain (in P). After the peptide bond, the chain passes to the tRNA in the A site,
- **Site E – Exit site:** it is where the unloaded tRNA (without amino acid) is expelled from the ribosome after transferring its amino acid.

Protein basic units: the amino acids

Codon: triplet of adjacent nucleotides in the mRNA chain that codes for a specific amino acid in the synthesis of a protein. The amino acids are the building blocks of the proteins. Starting from the start codon (i.e. AUG triplet of mRNA bases) each codon identifies a specific amino acid according to the rules in the Table.

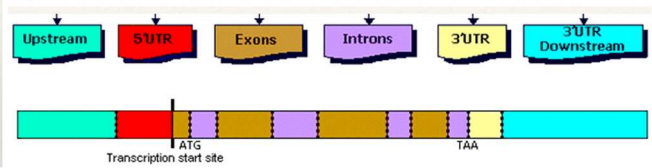
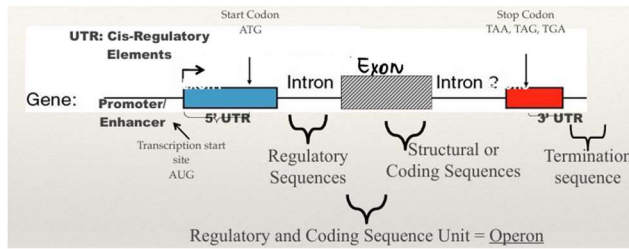
➔ The most terrible error is a bad translation with the result of a stop codon.

Amino Acids - Translation rule:

		Second base of codon				
		U	C	A	G	
First base of codon	U	UUU Phenylalanine UUC phe	UCU Serine UCC ser UCA ser UCG ser	UAU Tyrosine UAC tyr UAA STOP codon UAG STOP codon	UGU Cysteine UGC cys UGA STOP codon UGG Tryptophan	U C A G
	C	CUU Leucine CUC leu CUA leu CUG leu	CCU Proline CCC pro CCA pro CCG pro	CAU Histidine CAC his CAA Glutamine CAG gln	CGU Arginine CGC arg CGA arg CGG arg	U C A G
	A	AUU Isoleucine AUC ile AUA ile AUG Methionine (start codon)	ACU Threonine ACC thr ACA thr ACG thr	AAU Asparagine AAC asn AAA Lysine AAG lys	AGU Serine AGC ser AGA Arginine AGG arg	U C A G
	G	GUU Valine GUC val GUA val GUG val	GCU Alanine GCC ala GCA ala GCG ala	GAU Aspartic acid GAC asp GAA Glutamic acid GAG glu	GGU Glycine GGC gly GGA gly GGG gly	U C A G

➔ Different sequences encode the same amino acid.

Gene Structure



Untranslated region (UTR) – UTRs may affect gene regulation & expression. Normally untranslated. In Humans is partially translated because the start codon is part of the 5'UTR.

Cis-regulatory elements (CREs) – sequences that control the transcription of a gene. Cis-elements may be located in 5' or 3' untranslated regions or upstream the 5'UTR or within introns. Promoters facilitate the transcription of a particular gene and are typically upstream of the coding region (and the 5'UTR). (ex: promoter or enhancer).

Promoter – enhancer (regulatory elements):

- Regulatory sequence that controls the start of the transcription,
- Promoter: is where the RNA polymerase binds to initiate transcription,
- Enhancer: increases or decreases the efficiency of transcription.

5'UTR:

- Noncoding region,
- Contains the translation initiate site: ATG start codon,
- Regulate the translation process.

INTRON:

- Are present only into the eucaryote,
- Noncoding region, sometimes in RNA (noncoding RNA) acts like IF-ELSE, manage the transcription, *if somethings happened add the following part and translate,*
- Removed during the splicing.

EXON:

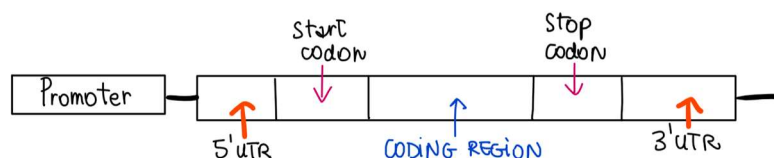
- Structural or coding sequence, contains information for proteins synthesis.

3' UTR:

- Noncoding region, contains termination sequence,
- Important for the stability and transport of mRNA.

➔ Starts: ATG

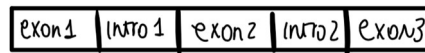
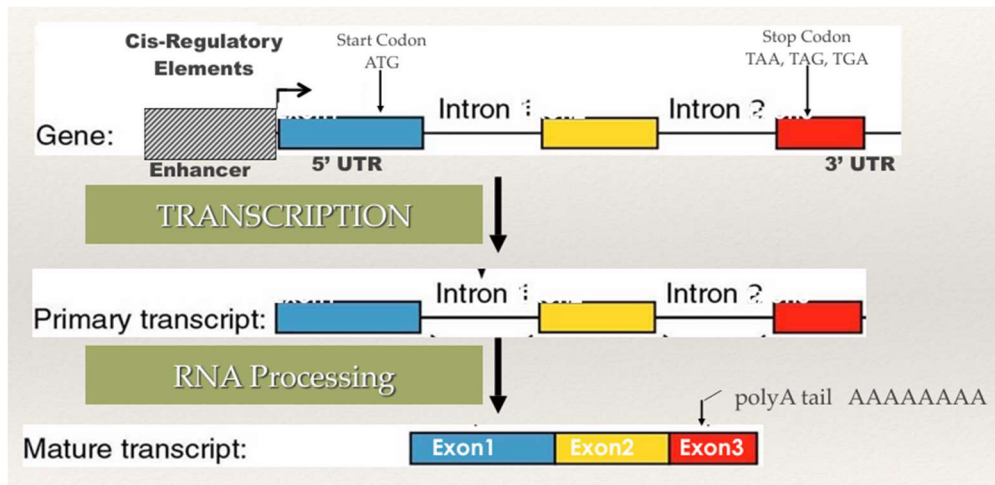
➔ Ends: TAA, TAG, TGA.



OPERON:

- Are present only into the procaryotic,
- Regulatory sequence + coding sequence.

Eukaryotic Gene Structure



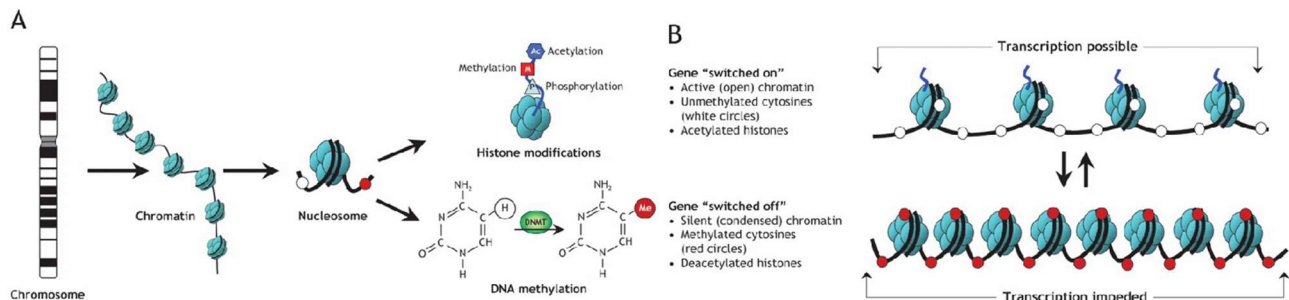
The gene is a Transcription Unit.

ORF → Open Reading Frame: new definition of Gene,

→ ORF = Gene but nobody refers to gene with ORF.

One gene → One protein [!!!!] (with possible several isoforms).

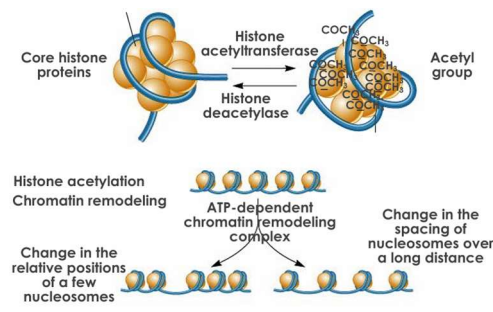
DNA Chromatin Structure



→ Methylation: to unzip the histones structure,

→ Nucleosome: histone + DNA,

○ Histones have the capability to stay very close to each other.

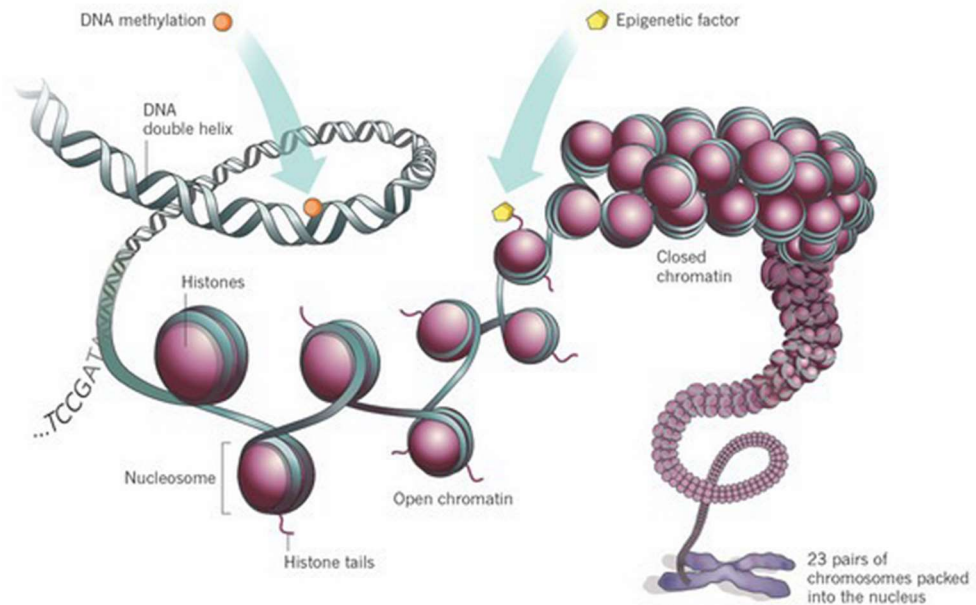


→ ATP: energy charged,

→ ADP: uncharged.

POLYGAMOUS DNA

DNA works with many partners. DNA methylation, for example, influences the way that genes are expressed without changing the underlying DNA sequence, and other epigenetic factors bind to histones to control when chromatin complexes open up and allow their DNA to be read.



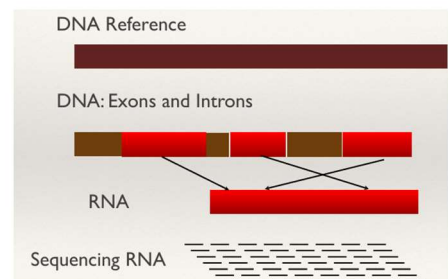
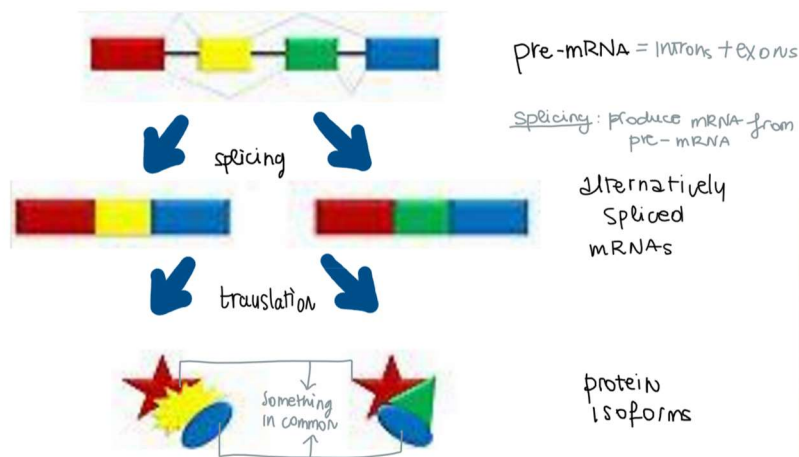
Alternative Splicing

Exons of the RNA produced by transcription of a gene are reconnected in multiple ways during RNA splicing. The resulting different mRNAs may be translated into different protein isoforms

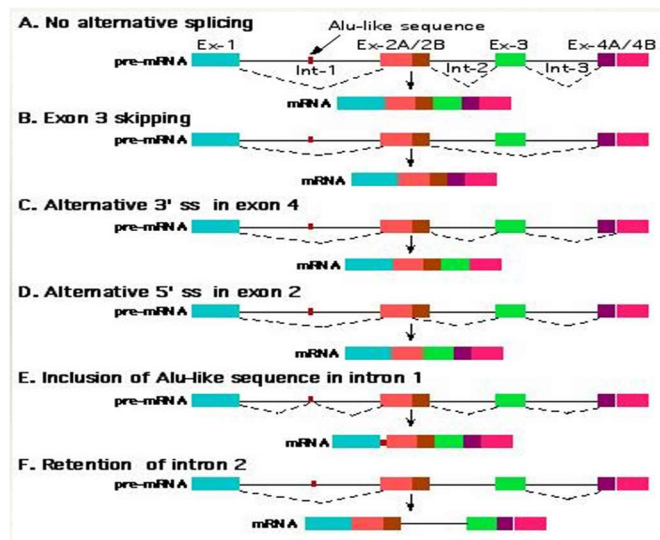
→ a single gene may code for multiple proteins.

It greatly increases the diversity of proteins that can be encoded by the genome; in humans, over 80% of genes are alternatively spliced.

There are numerous modes of alternative splicing observed. The most common is exon skipping. In this mode, a particular exon may be included in mRNAs under some conditions or in particular tissues and omitted from the mRNA in others.

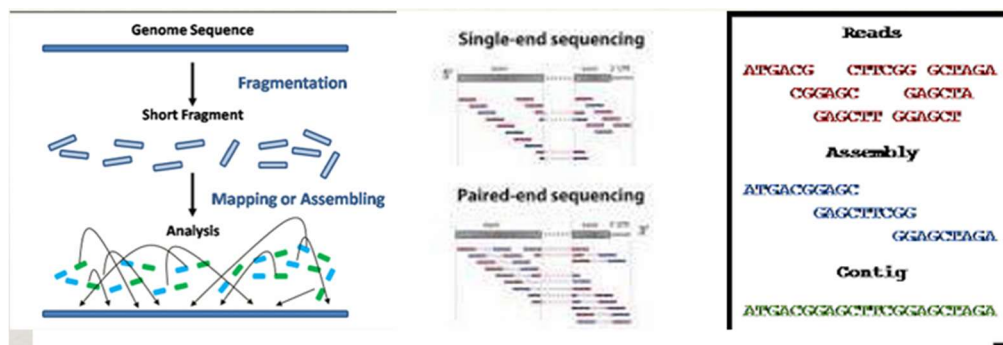


Examples of Alternative splicing:



- alternative splicing: remove all the introns and sometimes also some exons,
- splicing: remove all the introns and maintains the original sequence of exons.

Sequencing Splicing



Genome Sequencing Pipeline

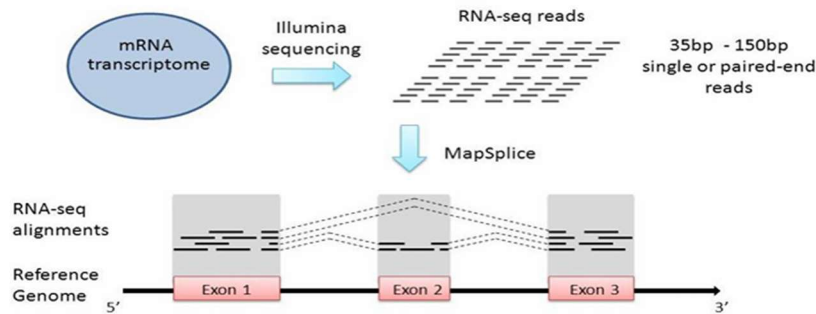
- **Genome Sequence:** starting from the complete genome,
- **Fragmentation:** the DNA is broken into small fragments (short reads),
- **Mapping or Assembling:**
 - o **Mapping:** if you have a reference genome, the fragments are mapped onto it,
 - o **Assembling:** if you do not have a reference, you look for overlaps between fragments to reconstruct the genome,
- **Analysis:** analysis of the results (mutations, expression, etc.).

Single-end vs Paired-end Sequencing

- **Single-end:** only one end of the fragment is read → less information.
- **Paired-end:** both ends of the fragment are read → greater precision and coverage → useful for identifying insertions, deletions or translocations.

Reads → Assembly → Contig

- **Reads:** the short sequences obtained (e.g. 100-150 bp).
- **Assembly:** overlapping reads are assembled into longer sequences.
- **Contig:** is a contiguous sequence reconstructed from multiple overlapping reads.



Step 1: RNA-seq

- Starting from the mRNA extracted from the cells (transcriptome),
- Sequencing is performed (e.g. Illumina) to obtain fragments (reads) from 35 to 150 bases,
- Reads represent the pieces of mRNA expressed in the various genes.

Step 2: Alignment with MapSplice

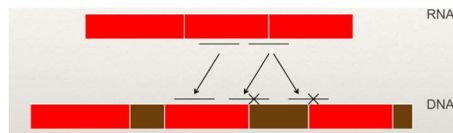
- MapSplice is an algorithm that aligns the RNA-seq reads to the reference genome,
- *Problem*: the mRNA is spliced, so the reads can cross multiple exons and skip introns (which do not appear in the mRNA),
- The map shows the reads that align within the exons or that cross different exons → useful for analyzing splicing.

Ex:

- If a read starts from Exon 1 and ends on Exon 2 → it indicates a splicing event between these two exons.
- This information is crucial for understanding which isoforms of a gene are expressed.

Splicing Effects on Alignment

Considering splicing enables a more precise read mapping and thus expression analysis.



Alternative slicing needs to take into account also all the possible combination between exons, in terms of order and number. A large number of reads across the real junctions would not be mapped without splicing-aware alignment. As an alternative there are tools (like BLAST) that trim the reads in small sequences and try to align them.

Note: the tools can work with DNA, with the RNA (DNA – introns) they can have problem of alignment.

Mutations

- By effect on **structure**:
 - **Small-scale mutations**, such as those affecting a small gene in one or a few nucleotides (e.g. point mutations, insertions, deletions, duplications),
 - **Large-scale mutations** in chromosomal structure (e.g. Gene duplication, Chromosomal translocation, Chromosomal inversions, Interstitial deletions),
- By effect on **function**:
 - **Loss-of-function** mutations are the result of gene product having less or no function. When the allele has a complete loss of function (null allele) it is often called an amorphic mutation. Phenotypes associated with such mutations are most often recessive, a function can be useful anymore,
 - **Gain-of-function** mutations change the gene product such that it gains a new and abnormal function. These mutations usually have dominant phenotypes,

➤ By effect on **fitness**:

- In applied genetics it is usual to speak of mutations as either harmful or beneficial.

Fusion Transcripts (Chimeras)

The fusion transcripts is a hybrid RNA molecule, which contains exons from different genes. These genes can be found on the same chromosome or on different chromosomes.

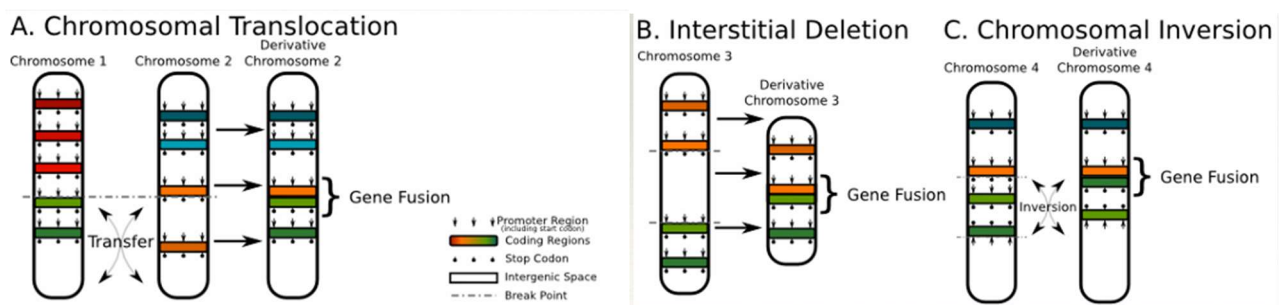
Chimeric RNA or fusion transcripts encoded by:

- a **fusion gene** at DNA level, exons come from different genes,
 - 2 genes become one, 2 functionality that look very similar collapse,
- a **fused mRNA** coming from multiple RNA polymerases,
 - There is no mutation in DNA,
 - When the RNA is affected by a pathology, some intron can be retained, and it can be part of the fusion too (as UTRs and other usually non coding regions).

A fusion gene is a hybrid gene formed from two previously separate genes. It can occur as the result of a translocation, interstitial deletion, or chromosomal inversion.

The "**normal**" (**cis**-)splicing processes a single molecule, in contrast, **trans-splicing results** in an RNA transcript that comes from multiple RNA polymerases on the genome. Exons from two different primary RNA transcripts are joined.

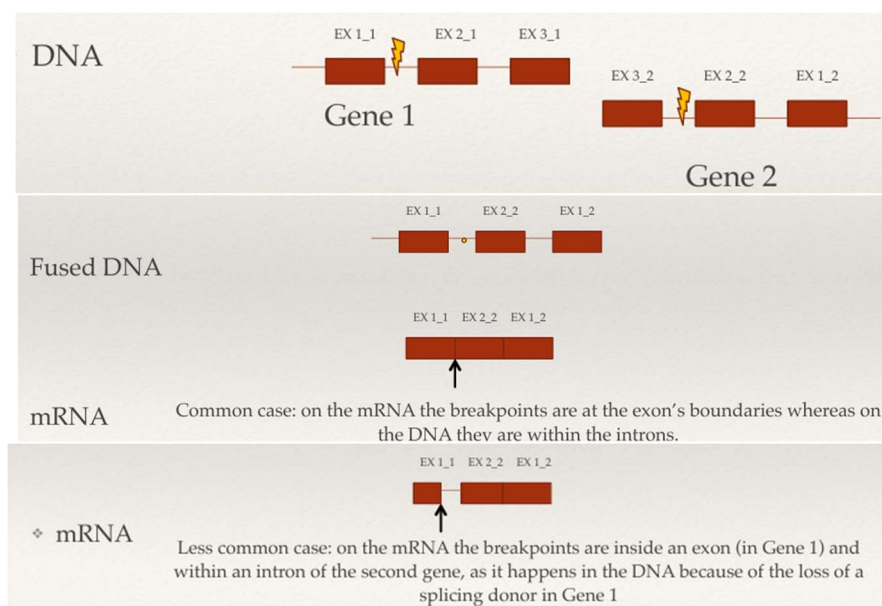
Gene fusion mechanisms



➔ not all the changes always bring improvements.

- A: new position of the gene,
- B: irrelevant gene are cut out,
- C: in the same chromosome new position of the gene (because works better, maybe).

Example of gene fusion in mRNA:



Computational Issues

Alternative splicing and chimerism makes analysis of RNA expression or alteration much more complex. The problem is to map the reads on the correct location on the genome avoiding mismatches and, vice versa, multiple matching. The identification of alternative splicing and chimerism it is very hard because splicing and fusion points are not known. In particular, fusion breakpoints are often not-canonical, i.e. genes can be broken inside exons, keeping subsequences of them and losing the others. Moreover, the genes involved in the fusion can be, often, located in different chromosomes and thus far away from each others.

Proteins & effects of mRNA aberrations on proteins

Importance of proteins:

- Main **catalysts** in biochemistry: **enzymes** (involved in virtually every biochemical reaction),
- **Structural components** of cells (both inside and outside of cells in tissues)
- **Regulatory functions** (e.g. gene expression, DNA replication, post-transcriptional regulation of mRNA, protein synthesis, etc.),
- Carrier and transport functions (ions, small molecules),
- Immune system activity (e.g. antibodies – produced by B cells – with complement proteins attach foreign invaders so other immune cells, such as T cells, can destroy them. T cells also secrete proteins known as cytokines. Cytokines work within a system of checks and balances to ensure the immune reaction does not go overboard),
- Mitosis and Meiosis processes,
 - o Mitosis: melting, sharing only ½ of the genome,
 - o Meiosis: cloning,
- DNA repair/tissue repair mechanisms,
- etc.

Impact of point mutations on protein's chain

A **point mutation**, or single base modification, is a type of mutation that causes a single nucleotide base change, insertion or deletion of the genetic material, DNA or RNA.

	Point mutations				
	No mutation	Silent	Nonsense	Missense	
				conservative	non-conservative
DNA level	TTC	TTT	ATC	TCC	TGC
mRNA level	AAG	AAA	UAG	AGG	ACG
protein level	Lys	Lys	STOP	Arg	Thr
					
				basic	polar

- ➔ some mutations don't bring issue and the translation result remains the same,
- ➔ other kind of errors bring some important and relevant issue, miss or wrong behaviour like unfunctional protein.

In frame or Frame-shifted

The term **frameshift mutation** indicates the addition or deletion of a base pair that implies a new sequence of triplets (then codons).

Mutation: deletion of a single base, then change of the reading frame.

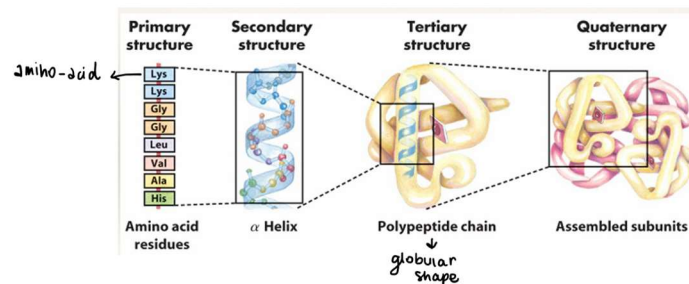
Deletion	Insertion
mRNA AAG AGG UUU AUA	mRNA AAG AGG UUA UA
protein Lys Arg Phe Iso	protein Lys Arg Leu Iso

- ➔ a complete shift can lose completely the meaning of all the following codons.

Protein folding: Levels of Protein Structure

- Primary Structure - amino acid sequence in a polypeptide, chain of amino acids created by the ribosomes,
- Secondary Structure - local spatial arrangement of a polypeptide's backbone atoms (α -helix and β -sheets), translated but not active,
- Tertiary Structure - three-dimensional structure of entire polypeptide,
- Quaternary Structure - spatial arrangement of subunits of proteins composed of multiple polypeptides (protein complexes).

Protein folding: Levels of Protein Structure



- ➔ Some proteins are functional already in tertiary structure,
- ➔ More complex proteins need also the quaternary structure.

Note: the structure influences the functionality.

Protein's Domains

Domains and their motifs/signature sequences constitute the content of many proteins secondary structures databases and are of enormous value in attempting to predict the function and structure of new proteins.

A domain is defined as sequentially consecutive amino acids that can fold up independently of other parts of the protein. The domain can evolve, function, and exist independently of the rest of the protein chain. The domain is the fundamental unit of protein structure characterizing protein's structure and function. A substantial proportion of all proteins are composed of more than one domain.

One domain may appear in a variety of different proteins. Molecular evolution uses domains as building blocks, and these may be recombined in different arrangements to create proteins with different functions. Inter-domain splicing, fusion, deletion, duplication and shuffling have occurred frequently during evolution, whereas intra-domain rearrangements have occurred rarely.

Bioinformatics

Alignment algorithms

➤ Short Read Alignment:

- Differences to conventional alignment:
 - millions of very short reads, rather than a few long ones, must be mapped to the genome,
 - mismatches can be, frequently, errors in the sequencing process. In some cases, they can be single point mutations/variations,
 - base-call quality information is more important, “that with 10% confidence”- the machine says that,
 - only small gaps are expected. A larger gap leads to a less biologically meaningful alignment.
 - mate-paired reads require special handling.

Alignment Tools

Type - FAMILY	Name	Link
General aligner	GMAP/GSNAP	http://research-pub.gene.com/gmap/
	BFAST	http://sourceforge.net/apps/mediawiki/bfast/index.php
	BOWTIE	http://bowtie-bio.sourceforge.net/index.shtml
	CloudBurst	http://sourceforge.net/apps/mediawiki/cloudburst-bio/index.php
	GNUmap	http://dna.cs.byu.edu/gnumap/index.shtml
	MAQ/BWA	http://maq.sourceforge.net/
	Perm	http://code.google.com/p/perm/
	RazerS	http://www.seqan.de/projects/razers.html
	Mrfast/mrsfast	http://mrfast.sourceforge.net/manual.html
	SOAP/SOAP2	http://soap.genomics.org.cn/
	SHRIMP	http://compbio.cs.toronto.edu/shrimp/
De Novo annotator	QPALMA/GenomeMapper/PALMapper	http://www.fiml.tuebingen.mpg.de/raetsch/suppl/palmapper
	SpliceMap	http://www.stanford.edu/group/wonglab/SpliceMap/
	SOAPs	http://soap.genomics.org.cn/
	G-Mo. R-Se	http://www.genoscope.cns.fr/externe/gmorse/
	TopHat	http://tophat.cbcb.umd.edu/
	SplitSeek	http://solidsoftwaretools.com/gf/project/splitseek
De Novo transcript assembler	Oases	http://www.ebi.ac.uk/~zerbino/oases/
	MIRA	http://sourceforge.net/apps/mediawiki/mira-assembler/index.php

- ➔ **General aligner** use a genome as reference,
- ➔ **De novo annotator** doesn't use a genome as reference, but they try create a new one by sequencing, no reference sequence. Analyse the genome and try to reconstruct DNA. More expensive in terms of computational efficiency-cost.

Examples of Alignment:

```

groan      colo-r      theatre     theatre
|||||      |||||      |||||::      |||||X
grown      colour     theater     theater
  ↓         ↘ mismatch
march

elephant   vermiform   vermiform----
|||:|     ::||:::     |||
eleg-ant   formation   ----formation
  ↓         ↘ deletion

disestablishment  disestablishment
|||      |  |||   |||      :|||
dis-----s-ent   dis-----sent
  
```

- the goal is to provide a score that brings some kind of information, for evaluate with is the best alignment,
- we want to analyse the difference, doesn't matter which is the reference sequence.

Relationship alignment-biology

- Query (Qry): string that we want to align,
- Subject (Sbj): reference string,

(Qry)	ACDEFG	ACDEFG	ACDEFG	AC--EFG
(Sbj)	ACDEFG	ACLEFG	AC--EFG	ACDEFG
Biological event	Conservation	Substitution	Insertion	Deletion
Alignment represent	Match	Mismatch	Gap	Gap

➤ Global Pair-wise alignment:

Example of a Score function:

ATCGATACG, ATGGATTACG		
ATCGAT-ACG ATGGATTACG		
Matches:	+1 +1 +1 +1 +1 +1 +1	= +8
Mismatches:	-1	= -1
Gaps:	-2	= -2
Total score = +5		
	Points for a matching letter:	1
	Points for a non-matching letter:	-1
	Points for inserting a gap:	-2

- Simplistic approach, in general difference mismatches have different scores.

Sequence similarity

Define the similarity of two sequences as the score of their highest-scoring (optimal) alignment.

- How do we find the optimal alignment of two sequence, and its score?
 - Depends on the different goals that we want to achieve,
 - Brute force enumeration is impractical, because the number of possible alignments becomes astronomically large for even short sequences.

Fortunately, the problem is soluble efficiently using a technique called dynamic programming.

Global Alignment and Dynamic programming

Dynamic programming is a method by which a larger problem may be **solved by first solving smaller**, partial versions of the problem. We demonstrate here how it may be applied to global sequence alignment, where at first, we are interested only in the similarity of two sequences, and not the alignment that yields this score.

A larger problem may be solved by first solving smaller, partial versions of the problem.

1)		Yj	G	A	C	T	A	C	
		Xi	A	C	G	C			
2)		Yj	G	A	C	T	A	C	
	(alternative pos)	Xi	-	A	C	G	C		
3)	(alternative pos)	Yj	-	G	A	C	T	A	C
		Xi	A	C	G	C			

→ X: query → i,

→ Y: subject → j,

→ $SIM(2,2)=SIM(1,2) \leftarrow SIM(x, y) = SIM(i, j)$

Definitions:

- $s(a, b)$ the substitution score for aligning letters a and b
- g the gap score for aligning any letter to a null
- X_i the partial sequence consisting of the first i letters of $X \equiv x_1 x_2 \dots x_m$
- Y_j the partial sequence consisting of the first j letters of $Y \equiv y_1 y_2 \dots y_n$
- $SIM(i, j)$ the similarity of X_i and Y_j

Consider the *last column* of an optimal alignment of X_i and Y_j . This column either aligns x_i to y_j , or x_i to a null, or y_j to a null. Because we do not allow "crossing", there are no other possibilities. This observation yields the following recurrence:

$$SIM(i, j) = \max \begin{cases} SIM(i-1, j-1) + s(x_i, y_j) & x_i \text{ and } y_j \text{ aligned} \\ SIM(i-1, j) + g & y_{j-1} \text{ aligned with a null} \\ SIM(i, j-1) + g & x_{i-1} \text{ aligned with a null} \end{cases}$$

In brief, we can solve for $SIM(m, n)$ by solving smaller versions of the problem first.

If X is the Query (Qry) string and Y is the Subject (Sbj), and i identifies the letters of X and j the letters of Y , $SIM(i, j) = SIM(i-1, j) + g$ implies a gap on X , the Qry.

In fact, in order to update the score, we compare the letter $(i-1)$ th of the Qry with letter j th of the Sbj. This is possible by adding a gap before the $(i-1)$ th letter on Qry sequence so that the letter $(i-1)$ th is in line with the j th letter on the Sbj.

1. First case (without a gap): the score is computed by comparing $(i-1)$ th letter on Qry with $(j-1)$ th letter on Sbj
2. Second case (adding a gap): the score is computed by comparing $(i-1)$ th letter on Qry with (j) th letter on Sbj. This implies a penalty due to the insertion of the gap.

BLAST - Basic Local Alignment Search Tool

It is a tool used to compare biological sequences with a dB of known sequences.

Primarily designed to identify homologous sequences:

- Blast is a hashed **seed-extend** algorithm,
 - o Define the threshold, this tool needs a lot of meta parameters,
- Functional conservation,
- Only some parts of a sequence are under positive selection,
- From a functional perspective, these are the interesting parts.

Finding seeds significantly increases the speed of BLAST compared to doing a full local alignment over a whole sequence. The choice of the seed is fundamental it needs to be not too long, short or too general.

BLAST first finds **highly conserved** or identical sequences which are then extended with a local alignment, so it's not very useful when you want a perfect match.

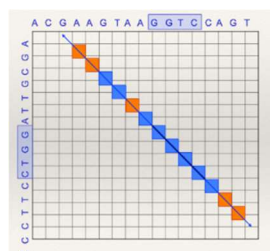
BLAST - seed and extend

Example: Seed size = 4, No mismatches in seed.

The matching word GGTC initiates an alignment. Extension to the left and right with no gaps until alignment score falls below 50%.

Output:

GTAAGGTCC
GTTAGGTCC



Speed of BLAST

Typically BLAST will take approximately 0.1 – 1 second to search 1 sequence against a database. Depends on size of database, e-value cutoff and number of hits to report selected.

➔ 60 million reads equates to 70 CPU days!

Even on multi-core systems this is too long! Especially if you have multiple samples! This is still true of FPGA and SIMD (vectorised) implementations of BLAST.

When NOT to use BLAST:

A typical situation: you have a piece of DNA sequence and want to extend it or find what gene it belongs to. In other words, you want an exact or near-exact match to a sequence that is part of an assembled genome. Short reads require very fast algorithms for finding near-exact matches in genomic sequences:

- BLAT: highly recommended, the BLAT paper (Kent WJ (2003) Genome Res 12:656-64) - it is famous for its unorthodox writing style,
- SOAP,
- Bowtie/Bowtie 2,
- MAQ,
- BWA,
- Shrimp2.

Adapted hashed seed-extend algorithms to work with shorted reads

Improve seed matching sensitivity:

- Allow mismatches within seed: BLAST,
- Allow mismatches + Adopt spaced-seed approach: ELAND, SOAP, MAQ, RMAP, ZOOM,
- Allow mismatches + Spaced-seeds + Multi-seeds: SSAHA2, BLAT, ELAND2,
- Above and/or Improve speed of local alignment for seed extension:
 - Single Instruction Multiple Data: Shrimp2, CLCBio,
 - Reduce search space to region around seed.

Suffix-Prefix Trie

Trie: tree structure that allow to store strings efficiently, faster string search.

A family of methods which uses a **Trie structure** to search a reference sequence:

- Bowtie,
- BWA,
- SOAP version 2,
- Trie – data structure which stores the suffixes (i.e. ends of a sequence),

Key advantage over hashed algorithms:

- Alignment of multiple copies of an identical sequence in the reference only needs to be done once,
- Use of an FM-Index to store Trie can drastically reduce memory requirements (e.g. Human genome can be stored in 2Gb of RAM),
- Burrows Wheeler Transform to perform fast lookups.

NEXT GENERATION SEQUENCING: Format and Tools

OUTLINE

DATA FORMATS:

- Fasta Format,
- MultiFasta Format,
- Fastq Format,
- SAM/BAM Format.

GENOME ANNOTATION FORMATS:

- Genome Annotation Concept,
- GFF Format,

- BED Format.

TOOLS FOR FILE ELABORATION:

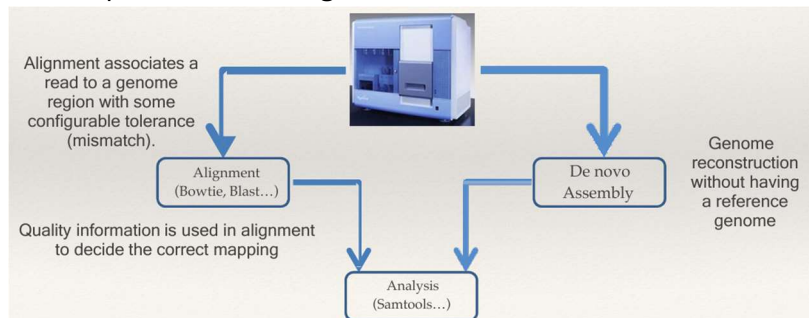
- SAMTools,
- BEDTools ALIGNMENT TOOLS,
- Sequence Alignment Concept,
- Bowtie,
- BLAST,
- TopHat.

ANALYSIS AND VISUALISATION TOOLS:

- SeqMonk.

Data Flow

Single paired-end reads are produced or using different kind of DIFFERENT OUTPUT FORMATS!



Data Formats

Fasta Format

FASTA format is a text-based (text-file) format for representing either **nucleotide sequences** or **peptide sequences**, in which nucleotides or amino acids are represented using **single-letter codes**. A sequence in FASTA format begins with a single-line description, followed by lines of sequence data. The **description line** is distinguished from the sequence data by a greater-than ("**>**") symbol in the first column. The word following the ">" symbol is the identifier of the sequence, and the rest of the line is the description (both are optional). It is recommended that all lines of text be shorter than 80 characters.

Description Line ← Sequence Data ←	<pre>>gi 5524211 gb AAD44166.1 cytochrome b [Elephas maximus maximus] LCLYTHIGRNIYYGSYLYSETWNTGIMLLITMATAFMGYVLPWGQMSFWGATVITNLFSAIPYIGTNLV EWINGGFSVDKATLNRFFAFHFILPFTMVALAGVHLTFLHETGSNNPLGLTSDSDKIPFHPYYTIKDFLG LLILILLLLLALLSPDMLGDPDNHMPADPLNTPLHIKPEWYFLFAYAILRSVPNKLGGVLAFLSIVIL GLMPFLHTSKHRSMMRLPLSQALFWTLTMDLLTLTWIGSQPVEYPYTHIGQMASILYFSIILAFPLIAGX IENY</pre>
--	---

➔ This example represents a proteins because there are more than 4 letters.

DESCRIPTION LINE: In the original Pearson FASTA format, one or more comments, distinguished by a semi-colon at the beginning of the line, may occur after the header.

SEQUENCE DATA: After the header line and comments, one or more lines may follow describing the sequence: each line of a sequence should have fewer than 80 characters. Sequences may be **protein sequences** or **nucleic acid sequences**, and they can contain gaps or alignment characters (some letters are used to identify the uncertainty). Sequences are expected to be represented in the standard IUB/IUPAC amino acid and nucleic acid codes, with these exceptions: lower-case letters are accepted and are mapped into upper-case; a single hyphen or dash can be used to represent a gap character; and in amino acid sequences, U and * are acceptable letters.

Nucleic Acid Code	Meaning	Mnemonic	Amino Acid Code	Meaning
A	A	Adenine	A	Alanine
N	A C G T U	aNy	B	Aspartic acid or Asparagine
M	A or C	bases with aMino groups	C	Cysteine
R	A or G if you don't know if A or G put R	puRine	D	Aspartic acid
W	A, T or U	Weak interaction	E	Glutamic acid
C	C	Cytosine	F	Phenylalanine
S	C or G	Strong interaction	G	Glycine
Y	C, T or U	pYrimidines → family	H	Histidine
G	G	Guanine	I	Isoleucine
K	G, T or U	bases which are Ketones	K	Lysine
-	gap of indeterminate length		L	Leucine
X	masked		M	Methionine
V	neither T nor U (i.e. A, C or G)	V comes after U	N	Asparagine
B	not A (i.e. C, G, T or U)	B comes after A	O	Pyrrolysine
D	not C (i.e. A, G, T or U)	D comes after C	P	Proline
H	not G (i.e., A, C, T or U)	H comes after G	Q	Glutamine
T	T	Thymine	R	Arginine
U	U	Uracil	S	Serine
			T	Threonine
			U	Selenocysteine
			V	Valine
			W	Tryptophan
			Y	Tyrosine
			Z	Glutamic acid or Glutamine
			x	any
			*	translation stop
			-	gap of indeterminate length

MultiFasta Format

MultiFasta format is an extension of Fasta Format.

A multiple sequence FASTA format or MULTIFASTA file would be obtained by concatenating several single sequence FASTA files.

→ Contains more than 1 fasta format information-structure.

```

Description Sequence 1 ← >SEQUENCE_1
                          MTEITAAVKELESTGAGMMDCKNALSETNGDFDKAVQLLREKGLGKAAKKADRLAAEG
                          LVSVKVSDDFITAAIRPSYLSYEDLDMTFVENYKALVAELEKENEERRRLKDPNKPEHK
Data Sequence 1 ←       IPQFASRKQLSDAILKEAEEKIKEELKAQKPEKIWDNIIPGKMNSFIADNSQLDSKLTIL
                          MGQFYVMDKKITVEQVIAEKEKEFGGKIKIVEFICFEVGEGLKKTEDFAAEVAAQL
Description Sequence 2 ← >SEQUENCE_2
                          SATVSEINSEIDFVAKNDQFIALTIDTTAHIQSNLSQVEELHSSTINGVKFEEYLYKSQI
Data Sequence 2 ←       ATIGENLVVRRFATLKAGANGVNGYIHTNGRVGVVIAAACDSAEVASKSRDLLRQICMH

```

→ More description lines and more sequence lines.

Fastq Format

FASTQ format is a text-based format for storing both a **biological sequence** (usually nucleotide sequence) and its corresponding **quality scores**. Both the sequence letter and quality score are encoded with a single ASCII character for brevity. A FASTQ file normally uses four lines per sequence.

```

@SEQ_ID
GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTGTTCACACTCACAGTTT
+
!''*((( (**+)) $$$++) ($$$$).1***-+*''))**55CCF>>>>>CCCCCCC65

```

- **Line 1** begins with a '@' character and is followed by a sequence identifier and an optional description.
- **Line 2** is the raw sequence letters.
- **Line 3** begins with a '+' character and is optionally followed by the same sequence identifier again.
- **Line 4** encodes the quality values for the sequence in Line 2.

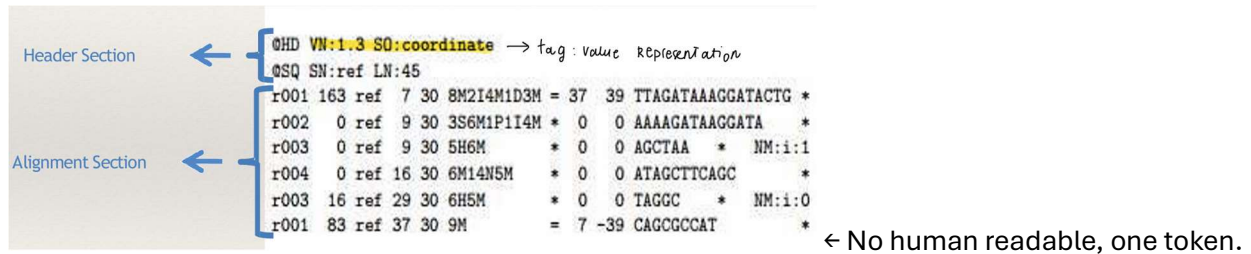
A quality value Q is an integer mapping of p (i.e., the probability that the corresponding base call is incorrect). Two different equations have been in use.

$$Q_{\text{sanger}} = -10 \log_{10} p$$

$$Q_{\text{solexa-prior to v.1.3}} = -10 \log_{10} \frac{p}{1-p}$$

SAM Format

SAM stands for Sequence Alignment/Map format; it is a **TAB-delimited text format** consisting of a header section, which is optional, and an alignment section. If present, the header must be prior to the alignments; header lines start with '@', while alignment lines do not. Each alignment line has **11 mandatory fields** for essential alignment information such as mapping position, and variable number of optional fields for flexible or aligner specific information.

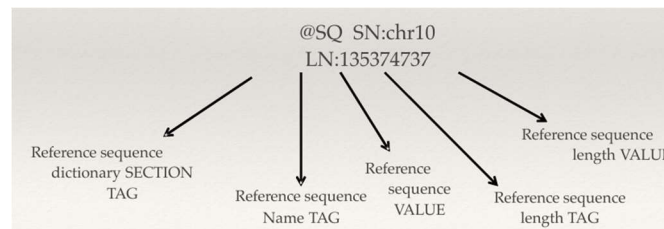


Header Section

Alignment Section

← No human readable, one token.

HEADER SECTION: Each header line begins with character '@' followed by a two-letter record type code. In the header, each line is TAB-delimited and each data field follows a format 'TAG:VALUE' where TAG is a two-letter string that defines the content and the format of VALUE.



Header Section Specifications:

Tag	Description
@HD	The header line. The first line if present.
VN*	Format version. <i>Accepted format:</i> /^[0-9]+\.[0-9]+\$/.
SO	Sorting order of alignments. <i>Valid values:</i> unknown (default), unsorted, queryname and coordinate. For coordinate sort, the major sort key is the RNAME field, with order defined by the order of @SQ lines in the header. The minor sort key is the POS field. For alignments with equal RNAME and POS, order is arbitrary. All alignments with '*' in RNAME field follow alignments with some other value but otherwise are in arbitrary order.
@SQ	Reference sequence dictionary. The order of @SQ lines defines the alignment sorting order.
SN*	Reference sequence name. Each @SQ line must have a unique SN tag. The value of this field is used in the alignment records in RNAME and PNEXT fields. Regular expression: [!-]+<->-] [!-]*
LN*	Reference sequence length. <i>Range:</i> [1,2 ²⁹ -1]
AS	Genome assembly identifier.
MS	MD5 checksum of the sequence in the uppercase, with gaps and spaces removed.
SP	Species.
UR	URI of the sequence. This value may start with one of the standard protocols, e.g http: or ftp:. If it does not start with one of these protocols, it is assumed to be a file-system path.
@RG	Read group. Unordered multiple @RG lines are allowed.
ID*	Read group identifier. Each @RG line must have a unique ID. The value of ID is used in the RG tags of alignment records. Must be unique among all read groups in header section. Read group IDs may be modified when merging SAM files in order to handle collisions.
CN	Name of sequencing center producing the read.
DS	Description.
DT	Date the run was produced (ISO8601 date or date/time).
FO	Flow order. The array of nucleotide bases that correspond to the nucleotides used for each flow of each read. Multi-base flows are encoded in IUPAC format, and non-nucleotide flows by various other characters. <i>Format:</i> /*! [ACMGRSVTWYHKDBN] +/
KS	The array of nucleotide bases that correspond to the key sequence of each read.
LB	Library.
PG	Programs used for processing the read group.
PI	Predicted median insert size.
PL	Platform/technology used to produce the reads. <i>Valid values:</i> CAPILLARY, LS454, ILLUMINA, SOLID, HELICOS, IONTORRENT and PACBIO.
PU	Platform unit (e.g. flowcell-barcode.lane for Illumina or slide for SOLID). Unique identifier.
SM	Sample. Use pool name where a pool is being sequenced.

PG	Program.
ID*	Program record identifier. Each PG line must have a unique ID. The value of ID is used in the alignment PG tag and PP tags of other PG lines. PG IDs may be modified when merging SAM files in order to handle collisions.
PN	Program name
CL	Command line
PP	Previous PG-ID. Must match another PG header's ID tag. PG records may be chained using PP tag, with the last record in the chain having no PP tag. This chain defines the order of programs that have been applied to the alignment. PP values may be modified when merging SAM files in order to handle collisions of PG IDs. The first PG record in a chain (i.e. the one referred to by the PG tag in a SAM record) describes the most recent program that operated on the SAM record. The next PG record in the chain describes the next most recent program that operated on the SAM record. The PG ID on a SAM record is not required to refer to the newest PG record in a chain. It may refer to any PG record in a chain, implying that the SAM record has been operated on by the program in that PG record, and the program(s) referred to via the PP tag.
VN	Program version
CD	One-line text comment. Unordered multiple CD lines are allowed.

ALIGNMENT SECTION: Each alignment line has 11 mandatory fields; these fields always appear in the same order and must be present.

Col	Field	Type	Regexp/Range	Brief description
1	QNAME	String	[!~?A-Z]{1,255}	Query template NAME
2	FLAG	Int	[0,2 ¹⁶ -1]	bitwise FLAG
3	RNAME	String	\[!~?A-Z\]+	Reference sequence NAME
4	POS	Int	[0,2 ²⁹ -1]	1-based leftmost mapping POSition
5	MAPQ	Int	[0,2 ⁸ -1]	MAPping Quality
6	CIGAR	String	\[!~?A-Z\]+	CIGAR string
7	RNEXT	String	\[!~?A-Z\]+	Ref. name of the mate/next segment
8	PNEXT	Int	[0,2 ²⁹ -1]	Position of the mate/next segment
9	TLEN	Int	[-2 ²⁹ ,2 ²⁹ -1]	observed Template LENgth
10	SEQ	String	\[!~?A-Z\]+	segment SEQUENCE
11	QUAL	String	[!~?A-Z]+	ASCII of Phred-scaled base QUALity+33

CIGAR STRING: It describes in a compact way how the reads are mapped to the reference genome; SAM is able to store clipped alignments, spliced alignments, multi-part alignments, padded alignments and alignments in colour space.

Op	BAM	Description
M	0	alignment match (can be a sequence match or mismatch)
I	1	insertion to the reference
D	2	deletion from the reference
N	3	skipped region from the reference
S	4	soft clipping (clipped sequences present in SEQ)
H	5	hard clipping (clipped sequences NOT present in SEQ)
P	6	padding (silent deletion from padded reference)
=	7	sequence match
X	8	sequence mismatch

➔ Provide the quality of the match, complex information.

CIGAR OPERATORS DESCRIPTION:

- M identifies a match or a mismatch in the alignment of a read to the reference.

Ref :	ATCTTAGTTTTCCTTTATGCTA	Cigar:	10M
Read:	GTTTTCCTT		

- I identifies an insertion to the reference.

Ref :	ATCTTAGTTTTCCTTTATG	Cigar:	3M 3I4M
Read:	TAGTTTTCCTT		5M 3I4M

- D identifies a deletion from the reference.

Ref :	ATCTTAGTTTAAACCTTTATG	Cigar:	3M 3D4M
Read:	TAGTTTTCCTT		5M

- N identifies a skipped region from the reference.

Ref :	ATCTTAGTTTAAACCTTTATGCTA	Cigar:	3M 3N4M
Read:	GTTTTCCTT		

- S identifies a soft clipped sequence in a read. A read may indeed not be aligned from the first residue to the last one.

Ref :	ATCTTAGTTTCCCTTTATGCTA	Cigar:	2S6M2S
Read:	caTTTCCcg		

- H identifies a hard clipped sequence in a read. It is similar to soft clipping.

They are different in that hard clipped subsequence is not present in the alignment record. The previous alignment can also be represented with CIGAR: 2H6M2H, but in this case, the sequence stored in SAM is TTTCCC, instead of CATTCCCCG if soft clipping is in use. P identifies a padding, a silent deletion from padded reference.

Ref :	ATCTTAGTTA**CCTTTATG	Cigar:	4M2P4M
Read:	TAGTTA**CCTT		

Genome Annotation Format

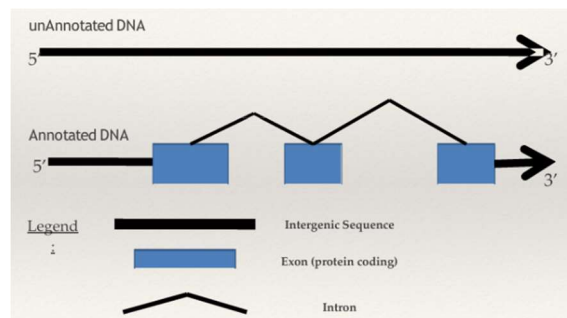
Genome annotation is the process of **attaching biological information** to sequences. It consists of two main steps:

1. Identifying elements on the genome, a process called gene prediction,
2. Attaching biological information to these elements.

2 main kinds of Annotations can be identified:

- **Structural annotation** consists of the identification of genomic elements: ORFs (Open Reading Frames) and their localisation; gene structure; coding regions; location of regulatory motifs.

Example of structural annotation:



- **Functional annotation** consists of attaching biological information to genomic elements: biochemical function; biological function; involved regulation and interactions; expression.

On-going projects relevant to genome annotation: ENCyclopedia Of DNA Elements (ENCODE); Entrez Gene; Ensembl; GENCODE; Gene Ontology Consortium; GeneRIF; RefSeq; Uniprot; Vertebrate and Genome Annotation Project (Vega). Gene annotation provided by Ensembl includes both automatic annotation (genome-wide determination of transcripts), and manual curation.

We need a format tracking:

- The location of the gene in the genome;
- The position of its exon-intron boundaries;
- The strand of DNA on which the gene lies;
- The source of annotation;
- The completeness of the gene structure.

GFF Format

GFF (General Feature Format) is a format for **locating** and **describing genes** and other localized features associated with DNA, RNA and protein sequences. GFF has several versions, the most recent of which is

GFF3. The GFF format consists of **one line per feature**, each containing **9 columns of data**, plus optional track definition lines. Fields contained in each of the line be tab-separated:

- **seqname**: name of the chromosome or scaffold,
- **source**: name of the program that generated this feature, or the data source,
- **feature**: feature type name,
- **start**: start position of the feature, with sequence numbering starting at 1; O end: end position of the feature, with sequence numbering starting at 1; O score: floating point value,
- **strand**: defined as + (forward) or - (reverse),
- **frame**: one of '0', '1' or '2'. '0' indicates that the first base of the feature is the first base of a codon, '1' that the second base is the first base of a codon, and so on,
- **attribute**: semicolon-separated list of tag-value pairs, providing additional information about each feature.

GFF3 file example:

ctg123	.	exon	1300	1500	.	+	.	ID=exon00001
ctg123	.	exon	1050	1500	.	+	.	ID=exon00002
ctg123	.	exon	3000	3902	.	+	.	ID=exon00003
ctg123	.	exon	5000	5500	.	+	.	ID=exon00004
ctg123	.	exon	7000	9000	.	+	.	ID=exon00005

Sequence Name	Feature	Start	End	Strand	Score	Frame	Attribute
---------------	---------	-------	-----	--------	-------	-------	-----------

A list of feature attributes in the format tag=value. Multiple tag=value pairs are separated by semicolons. Column 9 tags have predefined meanings:

ID	NAME	ALIAS	PARENT	TARGET	GAP	DERIVER-FROM	NOTE	DBXREF
1	2	3	4	5	6	7	8	9

- ➔ *: two exons in the same gene range, possible reasons:
- Alternative splicing,
 - Different annotation source,
 - Errors.

BED Format

BED Format is an alternative to GFF Format for **describing genomic features**. The BED format consists of one line per feature, each containing 3-12 columns of data, plus optional track definition lines. The first three fields in each feature line are required:

- **chrom**: name of the chromosome,
- **chromStart**: start position of the feature in standard chromosomal coordinates,
- **chromEnd** - End position of the feature in standard chromosomal coordinates.

chr1	213941196	213942363
chr1	213942363	213943530
chr1	213943530	213944697
chr2	158364697	158365864
chr2	158365864	158367031
chr3	127477031	127478198
chr3	127478198	127479365
chr3	127479365	127480532
chr3	127480532	127481699

Nine additional fields are optional:

- **name**: label to be displayed under the feature,
- **score**: a score between 0 and 1000,
- **strand**: defined as + (forward) or - (reverse),
- **thickStart**: field used by UCSC drawing code, not needed in Ensembl (set as 0 if using itemRgb field),

- **thickEnd:** as thickStart,
- **itemRgb:** an RGB colour value (e.g. 0,0,255).

chr7	127471196	127472363	Pos1	0	+	127471196	127472363	255,0,0
chr7	127472363	127473530	Pos2	0	+	127472363	127473530	255,0,0
chr7	127473530	127474697	Pos3	0	+	127473530	127474697	255,0,0
chr7	127474697	127475864	Pos4	0	+	127474697	127475864	255,0,0
chr7	127475864	127477031	Neg1	0	-	127475864	127477031	0,0,255
chr7	127477031	127478198	Neg2	0	-	127477031	127478198	0,0,255
chr7	127478198	127479365	Neg3	0	-	127478198	127479365	0,0,255
chr7	127479365	127480532	Pos5	0	+	127479365	127480532	255,0,0
chr7	127480532	127481699	Neg4	0	-	127480532	127481699	0,0,255

The remaining three fields - blockCount, blockSizes, and blockStarts - are used by the UCSC drawing code but not by Ensembl and will therefore be ignored.

Tools For File Elaboration

SAMTools

SAMtools is a set of utilities for interacting with and post-processing short DNA sequence read alignments in the SAM/BAM format, written by Heng Li. An API for each programming language is provided:

- BAMTools (C++);
- Picard (Java);
- Bio-SamTools (Perl);
- Pysam (Python).

SAMTools main commands are:

- **import:** SAM to BAM conversion;
- **view:** BAM to SAM conversion and subalignment retrieval;
- **sort:** sorting alignment;
- **merge:** merging multiple sorted alignment;
- **index:** indexing sorted alignment;
- **faidx:** FASTA indexing and subsequence retrieval;
- **tvview:** text alignment viewer.

BEDTools

The BEDTools utilities allow one to address common genomics tasks such as **finding feature overlaps** and **computing coverage**. The utilities are largely based on four widely-used file formats: BED, GFF, VCF, and SAM/BAM. Using BEDTools, one can develop sophisticated pipelines that answer complicated research questions by "streaming" several BEDTools together. Tools supported in the Suite.

Utility	Description
intersectBed	Returns overlapping features between two BED/GFF files. <i>Supports BAM format as input and output.</i>
pairToBed	Returns overlaps between a BEDPE file and a regular BED/GFF file. <i>Supports BAM format as input and output.</i>
pairToPair	Returns overlaps between two BEDPE files.
bamToBed	Converts BAM alignments to BED and BEDPE formats. <i>Supports BAM format as input.</i>
windowBed	Returns overlapping features between two BED/GFF files within a "window".
closestBed	Returns the closest feature to each entry in a BED/GFF file.
subtractBed	Removes the portion of an interval that is overlapped by another feature.
mergeBed	Merges overlapping features into a single feature.
coverageBed	Summarizes the depth and breadth of coverage of features in one BED/GFF file (e.g., aligned reads) relative to another (e.g., user-defined windows).
genomeCoverageBed	Histogram or a "per base" report of genome coverage.
fastaFromBed	Creates FASTA sequences from BED/GFF intervals.
maskFastaFromBed	Masks a FASTA file based upon BED/GFF coordinates.
shuffleBed	Permutes the locations of features within a genome.
slopBed	Adjusts features by a requested number of base pairs.
sortBed	Sorts BED/GFF files in useful ways.
linksBed	Creates an HTML links from a BED/GFF file.
complementBed	Returns intervals not spanned by features in a BED/GFF file.

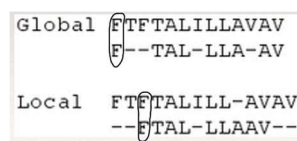
Alignment Tools

Sequence Alignment

A sequence alignment is a way of arranging the sequences of DNA, RNA, or protein to **identify regions of similarity** that may be a consequence of functional, structural, or evolutionary relationships between the sequences.

There are 2 categories of sequence alignment:

- **global alignments:** calculating a global alignment is a form of global optimization that "forces" the alignment to span the entire length of all query sequences. A general global alignment technique is the Needleman–Wunsch algorithm, which is based on dynamic programming,
 - entire DNA,
- **local alignments:** identify regions of similarity within long sequences that are often widely divergent overall. The Smith–Waterman algorithm is a general local alignment method also based on dynamic programming,
 - small portion of the DNA.



The choice of a specific alignment tool depends on:

- the sequencing technology, different tools produce different length results,
- the read length,
- the number of reads and available compute resources,
- the sensitivity/scoring requirements.

Note: short read allows increasing the number of miss match (GI vs GIANFRANCO → Gianni, Giovanni)

Bowtie

Bowtie is an **ultrafast** (very fast index searching), memory-efficient short read aligner geared toward quickly aligning large sets of short DNA sequences (reads) to large genomes.

Features:

- It aligns 35-base-pair reads to the human genome at a rate of 25 million reads per hour on a typical workstation;
- It indexes the genome with a Burrows-Wheeler index to keep its memory footprint small; for the human genome, the index is typically about 2.2 GB (for unpaired alignment);
- Multiple processors can be used simultaneously to achieve greater alignment speed.

Bowtie can also output alignments in the standard SAM format, allowing Bowtie to interoperate with other tools supporting SAM, including the SAMtools; Bowtie runs on the command line under Windows, Mac OS X, Linux, and Solaris. Bowtie is not a general-purpose alignment tool like BLAST. It **works best** when aligning **short reads** to large genomes, though it supports arbitrarily small reference sequences and reads as long as 1024 bases.

Bowtie is designed to be extremely fast for sets of short reads where:

- many of the reads have at least one good, valid alignment,
- many of the reads are relatively high-quality,
- the number of alignments reported per read is small.

Two main programs allow Botwie to perform the alignment:

4. **The bowtie-build indexer:** Bowtie-build builds a Bowtie index from a set of DNA sequences. It outputs a set of 6 files with suffixes .1.ebwt, .2.ebwt, .3.ebwt, .4.ebwt, .rev.1.ebwt, and .rev.2.ebwt. These files together constitute the index, all that is needed to align reads to that reference. The original sequence files are no longer used by Bowtie once the index is built. bowtie-build

```
bowtie-build [options]* <reference_in> <ebwt_base>
```

<reference_in> → Comma-separated list of FASTA files containing the reference sequences to be aligned.

<ebwt_base> → The basename of the index files to write.

[options] → A lot of options can be specified such as the input file format.

5. **The bowtie aligner:** Bowtie takes an index and a set of reads as input and outputs a list of alignments. Alignments are selected according to a combination of the -v/-n/-e/-l options which define which alignments are legal, and the -k/-a/-m/-M/--best/--strata options which define which and how many legal alignments should be reported.

Examples:

```
./bowtie -a --best -v 2 e_coli --suppress 1,5,6,7 -c ATCCCGGTCG
./bowtie -a --best --strata -v 2 --suppress 6,7 e_coli -c ATGCATC
./bowtie -a -m 3 -v 2 e_coli -c ATGCATCATGCGCCAT
```

BLAST

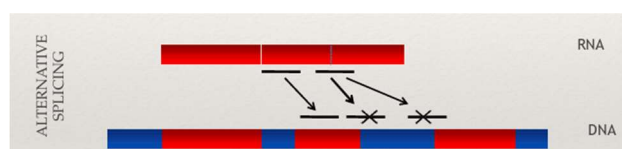
BLAST is an algorithm for comparing **primary biological sequence information**, such as the amino-acid sequences of different proteins or the nucleotides of DNA sequences. The standard input of blast is Fasta and this is why Fasta becomes so famous and used.

Features:

- BLAST is one of the most widely used bioinformatics programs, because it addresses a fundamental problem and the heuristic algorithm it uses is much faster than calculating an optimal alignment;
- Before fast algorithms such as BLAST were developed, doing database searches for protein or nucleic sequences was very time consuming because a full alignment was used (analyse all the entire set of possible permutation);
- The main idea of BLAST is that there are often high-scoring segment pairs (HSP) contained in a statistically significant alignment. BLAST searches for high scoring sequence alignments between the query sequence and sequences in the database using a heuristic;
- Input sequences are in FASTA Format whereas the output can be delivered in a variety of formats.

TopHat – splicing aware alignment tool

Alternative splicing and chimerism makes analysis of RNA expression much more complex: the problem is to map the reads on the correct location on the genome to make correct analysis.



A large number of reads across the real junction would not be mapped without splicing-aware alignment. TopHat is a program that aligns RNA-Seq reads to a genome in order to identify exon-exon splice junctions.

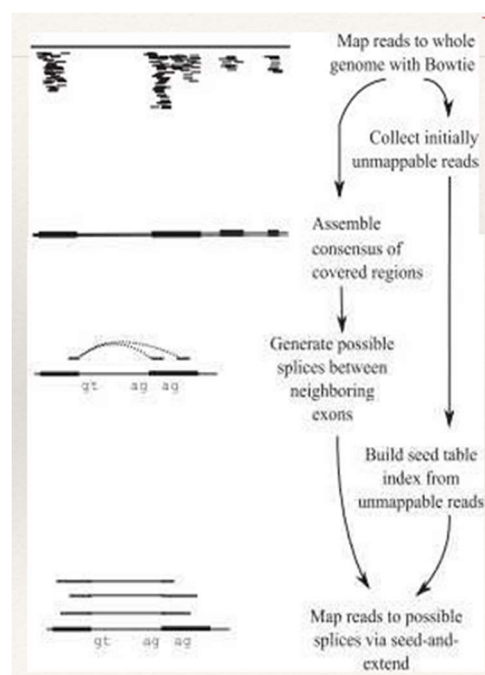
Features:

- It is built on the ultrafast short read mapping program Bowtie;
- It was designed to work with reads produced by the Illumina Genome Analyzer, although users have been successful in using TopHat with reads from other technologies;

- The software is optimized for reads 75bp or longer;
- The TopHat pipeline and its default parameter values are designed for detecting junctions even in genes transcribed at very low levels;
- The system may fail to detect junctions for a variety of reasons. The most common reason for missing a junction is that the transcript has very low sequencing coverage;
- Junctions spanning very long introns or short introns with non-canonical donor and acceptor sites (such as GC-AG introns) will also be missed;
 - o No pattern can help the machine to be sure
- It takes as input a comma-separated list of files containing reads in FASTQ or FASTA format;

RNA-Seq reads are mapped against the whole reference genome, and those reads that do not map are set aside. An initial consensus of mapped regions is computed by Maq. Sequences flanking potential donor/ acceptor splice sites within neighboring regions are joined to form potential splice junctions. The reads are indexed and aligned to these splice junction sequences.

- *accepted_hits.bam;*
- *junctions.bed;*
- *insertions.bed.*



Analysis and Visualisation Tools

SeqMonk

High throughput sequencing machines have the ability to generate tens of millions of sequences reads in a single run, containing hundreds of millions of bases of sequence: programs designed with high-throughput sequencing in mind have to be used. SeqMonk is a program developed at the Babraham Institute which was designed to help with the analysis of **mapped sequencing data**. It combines an **annotated genome viewer** with a **data viewer** allowing to visualise the data against an annotated genome. It provides tools for quantitating and filtering data and provides an environment in which it is easy to explore the data and try out different analyses in real time.

CONFIGURING SeqMonk: The current state of the installation can be seen in a panel of information when SeqMonks first starts up.

GREEN ICON	→	pass
BLUE ICON	→	information
RED ICON	→	attention (configuration options, updates...)

DISK CACHES:

- Default mode: store all data in memory. Quick operations, but limits on how much data can you put into one project,
- Disk based caching option: write out sequence read and genome annotation data to the hard drive when it is not being used. This option allows to load 20X as much data into SeqMonk at the expense of the program pausing briefly when it needs to load data back into memory from the disk.

GENOMES FOLDER:

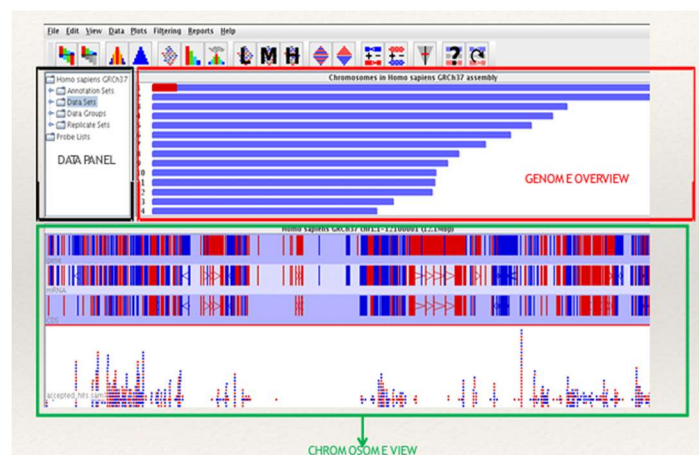
- Default mode: use pre-built genome annotation available in SeqMonk installation,
- Genome option: download the prefer genome and specify its directory.

EXTRA ANNOTATION Additional information can be loaded from different file formats. The new features will be immediately available in the genome view.

DATA FORMATS The information necessary to import data is a series of mapped genomic positions (chromosome, start, end positions, strand). Different data formats are supported (TXT, GFF, BED, SAM, BAM...)

For any standard format a couple of options are offered, such as:

- Single/Paired-end import;
- Remove Duplicates;
- Extending single end reads;
- Spliced import.

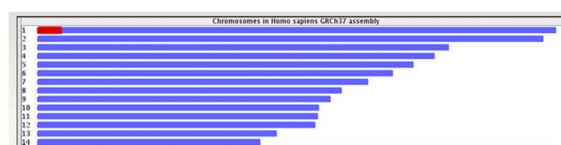


➔ Red: top strand, blue: bottom strand.

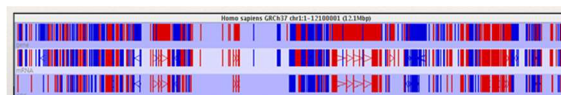
GROUPING SAMPLES Row data can be grouped together into larger groups:

- Data Groups,
- Replicate Set.

GENOME OVERVIEW It shows a graphical representation of the whole genome laid out in chromosomes; The red box on a specific chromosome indicates a region currently visible in the chromosome view shows a graphical representation of the whole genome laid out in chromosomes; Clicking and dragging anywhere on the genome view the chromosome view will be moved to show the selected region.



CHROMOSOME VIEW It is the most detailed view of the genome: It is arranged into tracks.



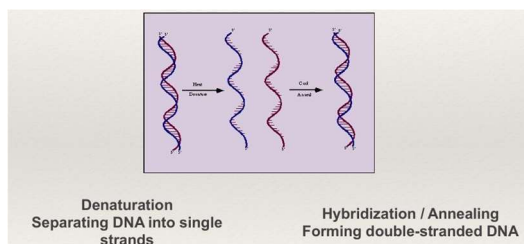
The tracks with blue backgrounds contain annotation information for different classes of features (depends on which feature you want to annotate). These appear as blocks coloured according to their orientation. If a feature has linked locations the blocks are joined.

Each white/grey data track contains the data for one data set or data group. Both raw read positions and quantitated values calculated from the reads can be shown.



NGS TECHNOLOGY

DENATURATION/HYBRIDIZATION



NUCLEIC ACID THERMODYNAMICS SINGLE 2 DUPLEX

- **Hybridization** is the process of establishing a non-covalent, sequence-specific interaction between two or more complementary strands of nucleic acids (i.e. DNA/RNA) into a single hybrid, which in the case of two strands is referred to as a duplex,
 - o It includes annealing as an initial phase, but is often longer and more controlled,
- **Annealing**, in genetics, means for DNA or RNA to pair by hydrogen bonds to a complementary sequence, forming a double-stranded polynucleotide. The term is often used to describe the binding of a DNA probe, or the binding of a primer to a DNA strand during a polymerase chain reaction (PCR). The term is also often used to describe the reformation (renaturation) of complementary strands that were separated by heat (thermally denatured),
- **DNA denaturation**, also called DNA melting, is the process by which double-stranded deoxyribonucleic acid (DNA) unwinds and separates into single-stranded strands through the breaking of hydrogen bonding between the bases. Both terms are used to refer to the process as it occurs when a mixture is heated, although "denaturation" can also refer to the separation of DNA strands induced by chemicals.

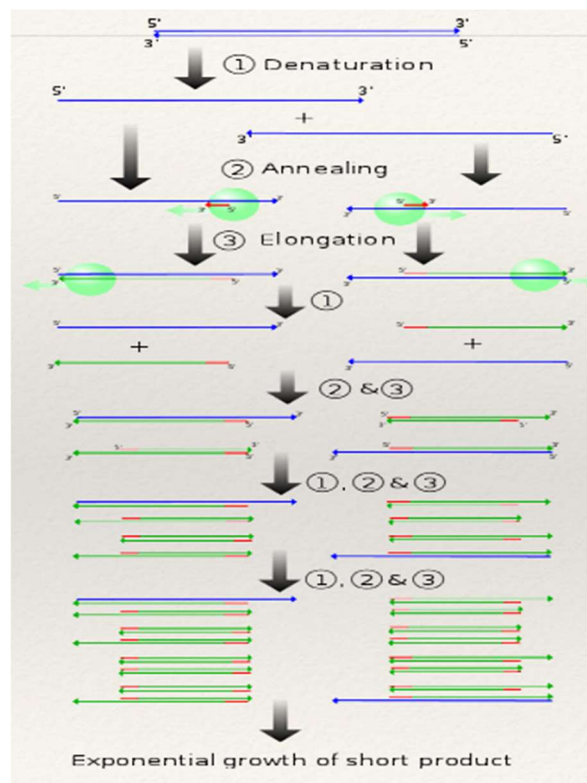
For multiple copies of DNA molecules, the melting temperature (T_m) is defined as the temperature at which half of the DNA strands are in the double-helical state and half are in the random coil or single-stranded state.

POLYMERASE CHAIN REACTION (PCR)

The polymerase chain reaction (PCR) is a technique in molecular biology to **amplify** a single or a few copies of a **piece of DNA** across several orders of magnitude, generating thousands to millions of copies of a particular DNA sequence. The method relies on thermal cycling, consisting of cycles of repeated heating and cooling of the reaction for DNA melting and enzymatic replication of the DNA.

1. Denaturing at 94–96 °C,
2. Annealing at ~65 °C,
3. Elongation at 72 °C. At this step the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding bases that are complementary to the template in 5' to 3' direction, condensing the 5'-phosphate group of the bases with the 3'-hydroxyl group at the end of the nascent (extending) DNA strand. Exponential growing.

Four cycles are shown here. The **blue lines represent the DNA template** to which primers (red arrows) anneal. These are extended by the DNA polymerase (light green circles), to give shorter DNA products (green lines), which themselves are used as templates as PCR progresses.

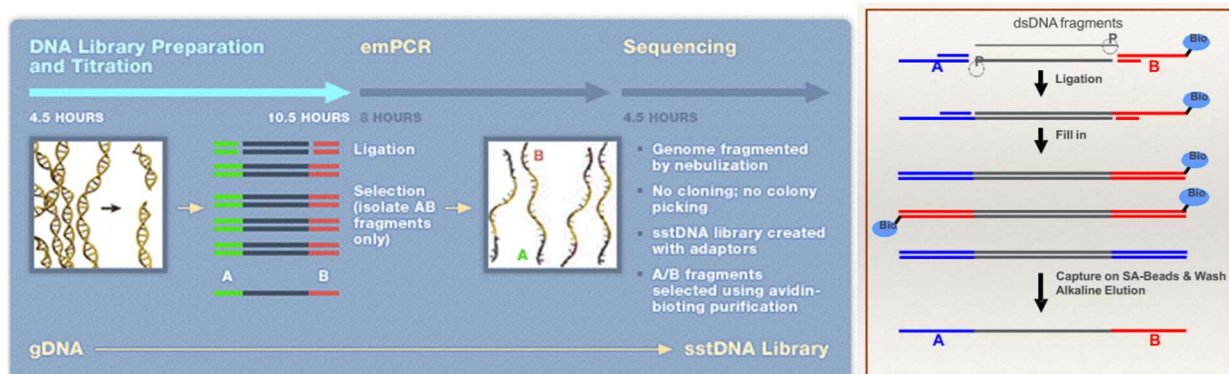


DNA LIBRARY PREPARATION

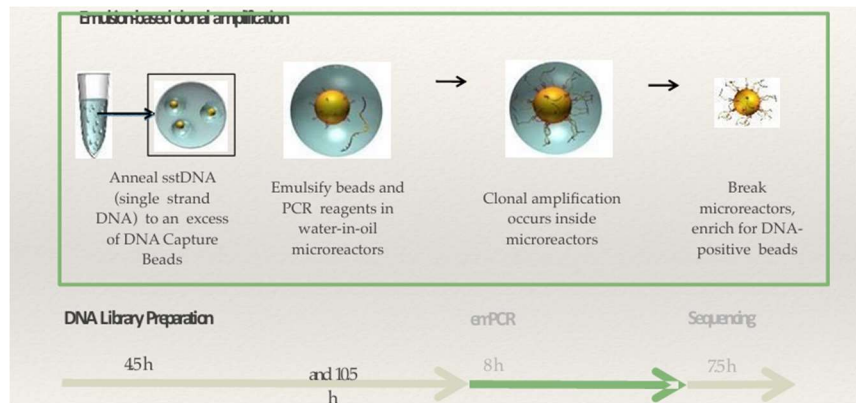
Most sequencing approaches use an in vitro cloning step to **amplify individual DNA molecules** that is because their molecular detection methods are not sensitive enough for single molecule sequencing.

Preparation of the DNA library consists of a few simple steps:

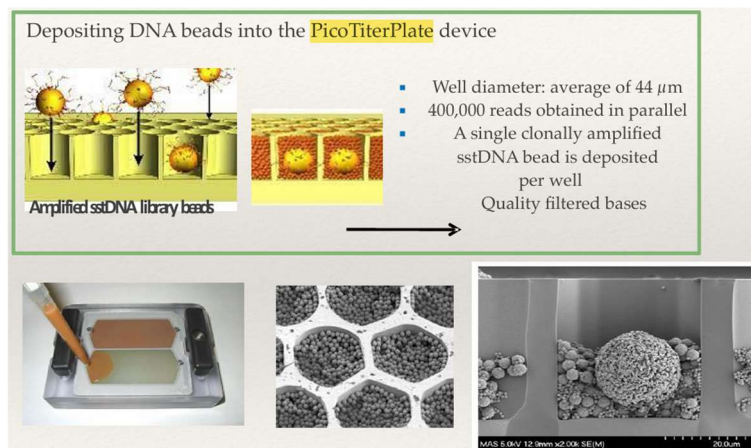
1. **Genomic DNA is fractionated into smaller fragments** (300-500 base pairs),
2. that are subsequently **phosphorylated** (addition of a phosphate (PO₄) group; phosphorylation activates or deactivates many protein enzymes),
3. **Short Adaptors (A and B) are then ligated onto the ends of the fragments**. These adaptors provide priming sequences for both amplification and sequencing of the sample-library fragments. Adaptor **B** contains a 5'-**biotin** tag that enables immobilization of the library onto streptavidin coated beads (in fact, Biotin binds very tightly to the tetrameric protein avidin (also streptavidin) creating one of the **strongest known protein-ligand interactions**, approaching the covalent bond in strength; biotin-binding is resistant to extremes of pH, temperature, organic solvents, denaturants, detergents and proteolytic enzymes).



1) CLONAL AMPLIFICATION DURING EMPCR

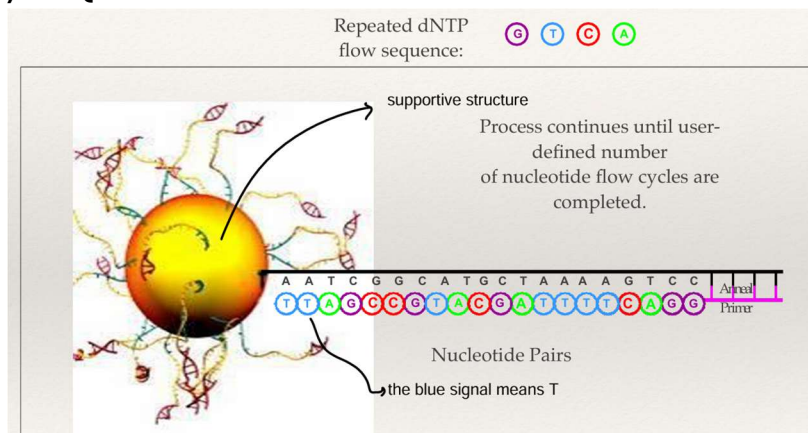


2) SEQUENCING - BEAD LOADING ON PICO TITER PLATES



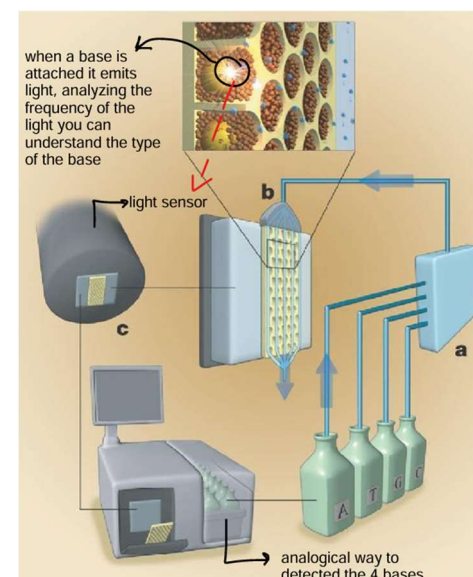
➔ All the pieces of DNA are attached on the ball so you can easily move the balls and put them in the hole.

3) SEQUENCING-BY-SYNTHESIS



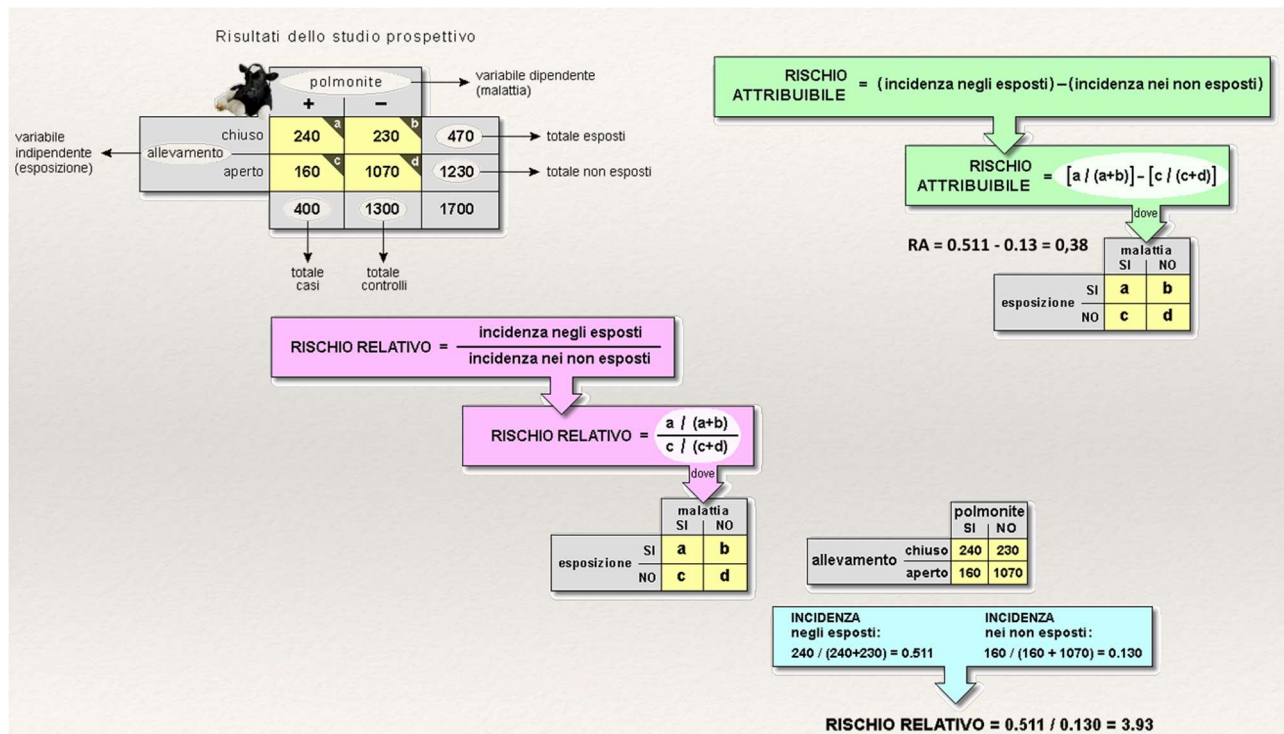
The sequencing instrument consists of the following major subsystems:

- a fluidic assembly
- a flow chamber that includes the well containing fibre-optic slide
- a CCD camera-based imaging assembly
- a computer that provides the necessary user interface and instrument control.



Data and Statistics

Know your Enemy: Be careful not to confuse **relative risk** with **attributable risk**. The two measures have different purposes. The RR is a measure of the **strength** of the association between the **disease factor** and the **disease itself** and is a very useful index on the road to demonstrating that the factor is the cause of the disease. The AR, on the other hand, represents **a measure of the impact** that the factor under consideration has on the population under study.



Statistics - The glitch

		Disease		iR	AR
		YES	NO		
Drug A	Exposure YES	1	230	0,0043	0,0018
	Exposure NO	1	400	0,0025	
Drug B	Exposure YES	2	230	0,0086	0,0061
	Exposure NO	1	400	0,0025	

		Disease		iR	AR
		YES	NO		
Drug A	Exposure YES	3	2300	0,00130	0,00010
	Exposure NO	3	2500	0,00120	
Drug B	Exposure YES	4	2300	0,00174	0,00054
	Exposure NO	3	2500	0,00120	

Statistics

Statistics is intended to explore the real world through the collection, analysis and interpretation of data. It allows the analysis of numerical data in which there is great variability.

Does proximity to high voltage networks cause disease X?

Due to the great variability inherent in the medical-biological material:

- genetic variability between individuals,
- variability due to the growth and development of individuals,
- variability of the reactions of the same individual at different times, or of the measurements performed on the same individual at different times.

Variability

Statistics help in the quantitative study of variability, allowing us to understand, manage and reduce variability. The variability of the phenomenon is studied by carrying out measurements or introducing qualitative parameters. We can hypothesise the law of the phenomenon, that is, the mathematical relationship between data for which a connection is sought, (eg. Height of an individual and length of the spine).

➔ Statistical Inference based on Probability.

TODO List

- Identify the aims of the research.
- Planning the research in order to study the problem for which an answer is sought.
- How to get the data you need?
- What statistical method to use to analyze them?
- How to interpret the results?
- How to present the results?

Data Collection is composed by, where the data come from:

- ➔ Survey, trust the perception of the people, different personal scale,
- ➔ Experiments,
- ➔ Field Observation,
- ➔ longitudinal studies.

Sampling

The **sampling design** or sampling **plan** is the method used to select the sample. The sample must be chosen randomly so as not to favor the inclusion of some elements over others, or the self-selection among those who have to answer.

- ➔ N. B. Choosing "randomly" does not mean "randomly",
- ➔ Sample set is hard to create and it's necessarily that is representative.

Choosing a sample in a random way (random) is given, to all elements, the same probability of being chosen. The choice of data can be complicate due to the fact that we don't know which could be the interesting-relevant data.

Data exploration

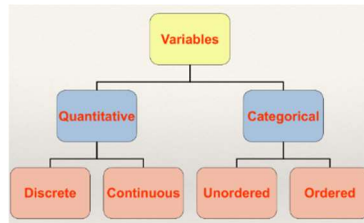
- Graphs: histograms, boxplots, ...
 - o The graphs help to represent all the data into a single picture,
- Numerical: summaries mean, dispersion, percentiles, ...

Through the exploratory analysis of the data, we try to understand what SHAPE they have.

Variables

Variables can be:

- **Qualitative** variable measures or matters something. (the height of an individual, the number of children in a family),
- **categorical** variable, defines the group to which it belongs. Eg. blood group, sex.



Categorical Variables

Categorical variables (with nominal scale) can be ordered or unordered:

- **unordered** variables cannot be sorted Eg. the species, sex, type of habitat,
- **Ordered** variables (with ordinal scale) that can be ordered Eg. the scale of abundance for the classification of the abundance of different plant species (dominant, abundant, frequent, uncommon, rare).

Quantitative Variables

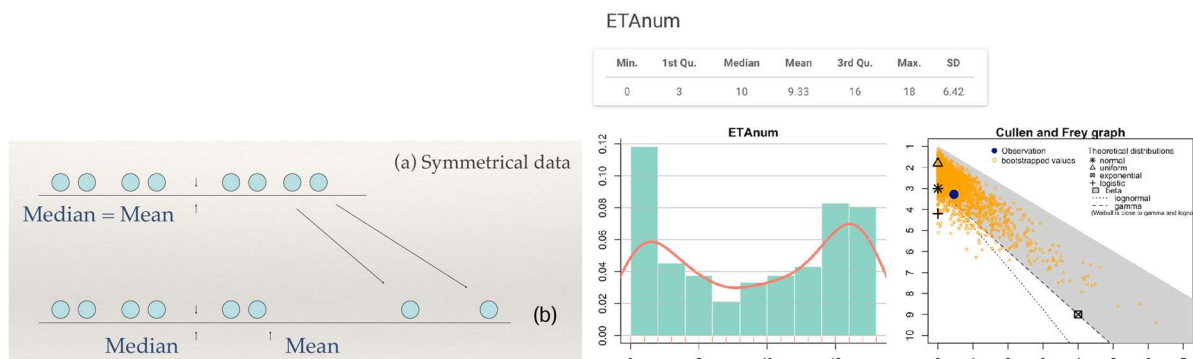
- **Discrete** quantitative variables Eg. the number of eggs laid by a bird,
- **Continuous** quantitative variables Eg. the temperature, the weight of an individual.

It is important to specify the level of measurement of the observed data to determine the statistical procedure to be used to analyze them.

Are there any outliers or gaps in the values? Is the data symmetrical? Is data distribution unimodal? Where is the distribution center? Is the data highly variable?

➔ Experience is necessary to answer this questions, but at the same time knowing the answers is necessary to work well and understand the result of our work.

- **Position indices:** mean, median, mode, quartiles, percentiles,
- **Indices of variability or dispersion:** range of variation, variance,
- **Shape indices:** kurtosis, index of symmetry.

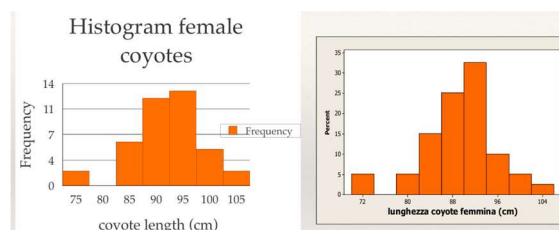


(b) The two observations with the largest value shifted to the right (skewed/asymmetrical data).

Charts

What does the graph reveal about the main characteristics of the data examined? The chart helps **interpret** the data. Pay attention to the choice of the graph. The graphical representations are numerous and must be chosen in relation to the types of variables and the scales used.

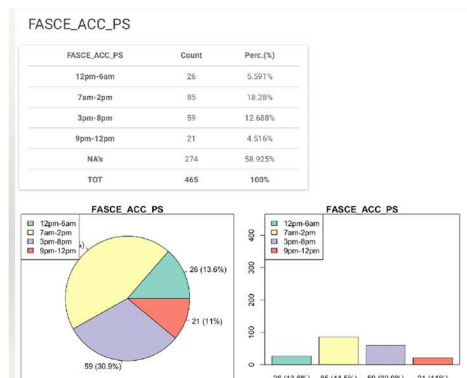
Ex: Categorical Variables



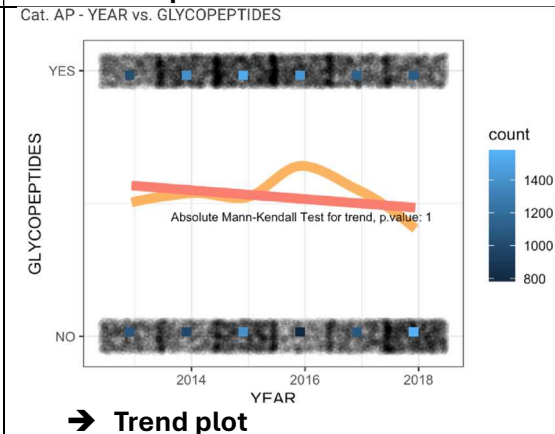
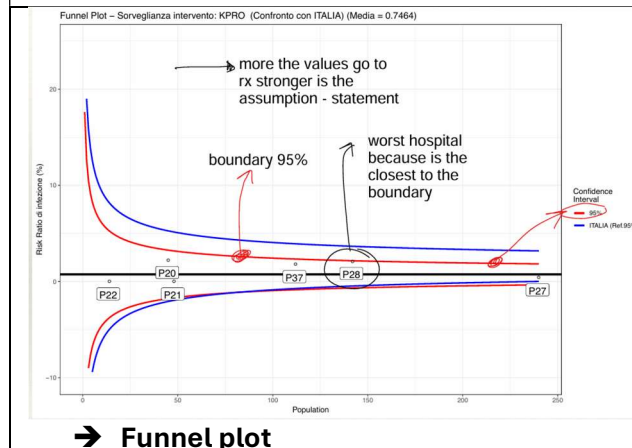
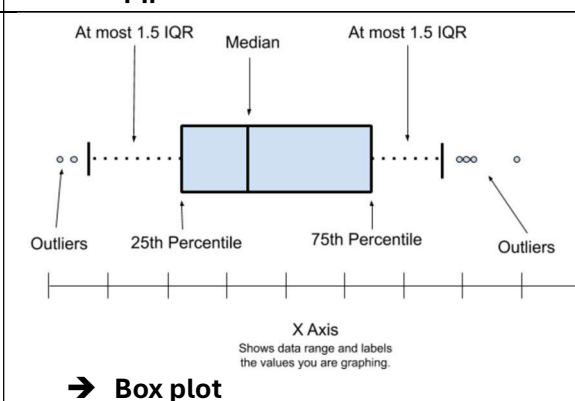
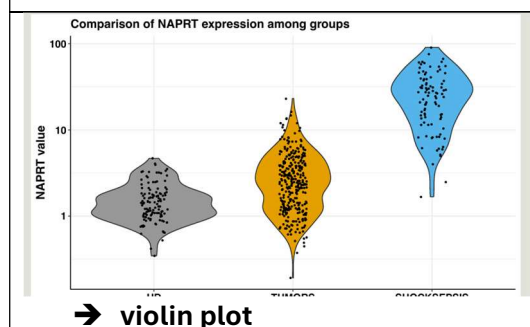
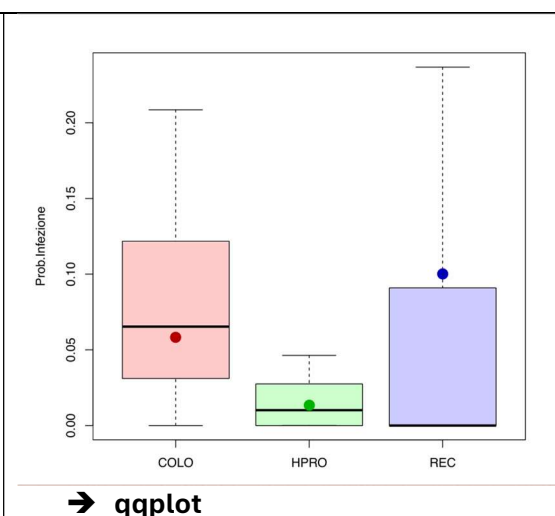
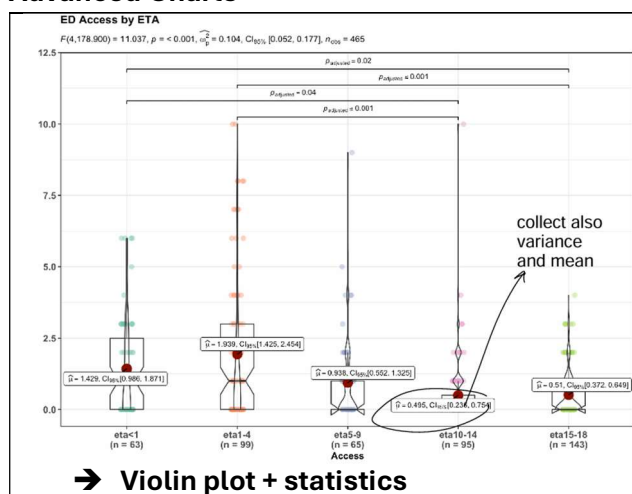
Pay attention to the choice of classes. Different class choices can lead to different looking histograms.

Suggestions:

- An appropriate histogram should have between 5 - 20 classes,
- Histograms are used when the sample size n is large enough,
- The histograms show: the form of distribution, outliers, grouped values, breaks in values.



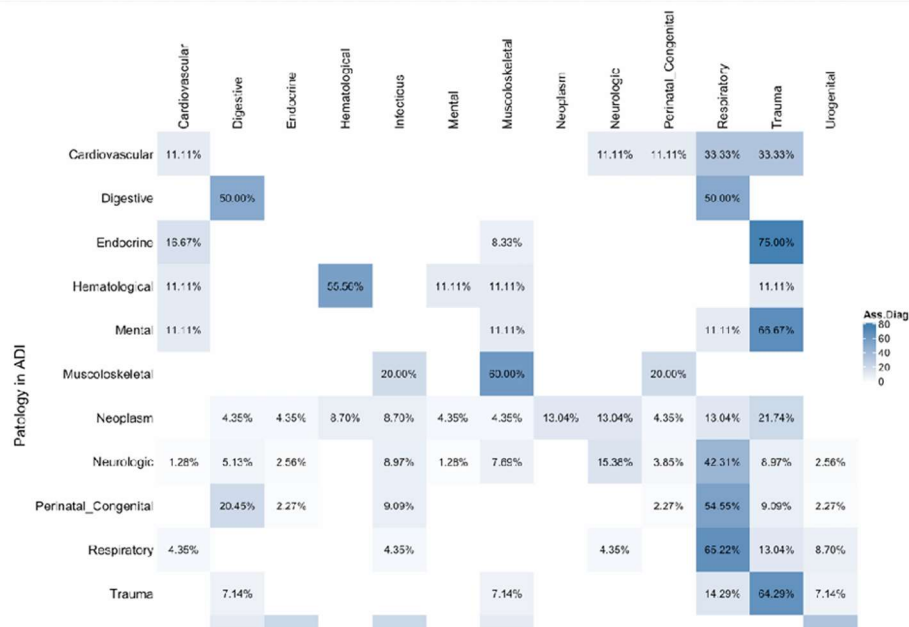
Advanced Charts



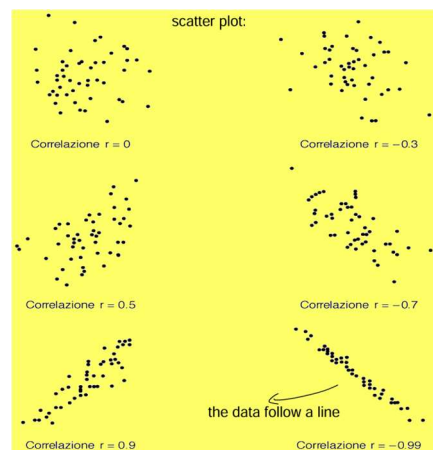
Variable's Relations

Regression line: Statistical model that describes the linear relationship between two quantitative variables. Describes how a response variable y changes (linearly) when the explanatory variable x changes, it is often used to predict new values of y from new values of x , determines how much of the variability (uncertainty) of y can be explained by the linear relationship to x , and how much of this variability remains unexplained.

Heat map: Graphical representation of data, describe general trends and any deviations from these trends, if the general trend is fairly regular, use a model (mathematical, statistical) to describe this trend. It can represents both relation between columns either auto-correlation.



Correlation:



Causality (!!!)

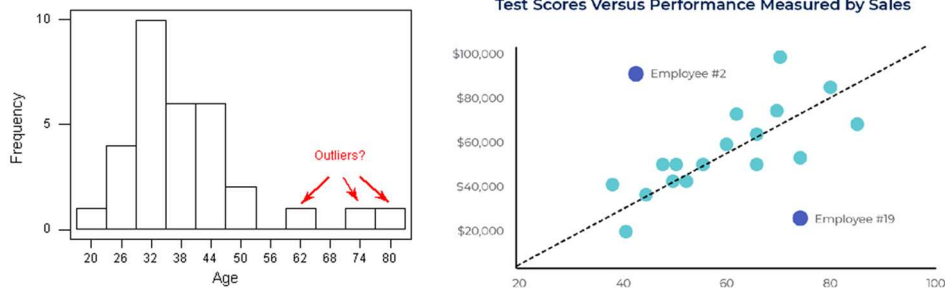
An association between an explanatory variable x and a response y , even if very strong, does not necessarily imply that y is modified by x .

Example: there is a high positive correlation between the number of televisions per person x and the average life span y for different nations of the world, in countries with many televisions, people live longer (the wealth of a nation can be a hidden variable).

In general, it is advisable to be cautious in stating that there is a cause-effect relationship between 2 variables, just because they are strongly associated. **A strong correlation does not necessarily imply a causal relationship.**

Outlier

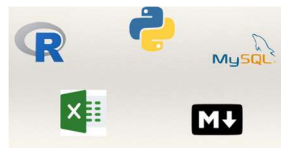
An outlier is an observation that does not follow the general model assumed by most of the observations. The points that, looking at a scatter plot, have high residuals can be considered outliers. An observation is influential if, eliminating it, it would profoundly change the result. In a scatter plot, the points that can be considered outliers in the x direction are often influential points in the determination of the least squares regression line.



TODO List

- Try to find a general trend for the observed data,
- Identify any deviations from this trend,
- Identify the most significant characteristics/features of the data distribution such as shape, center, and dispersion,
- Identify any outliers. They can be interesting / unusual mistakes or observations.

Tools



Techniques

Analytical techniques (statistical or computational) that are used to:

- Describe data (univariate, multivariate statistics),
- Estimate relationships between variables (GLM, fixed effects),
- Classify or predict events (ML, DL, graph theory),
- Make global inferences (meta-analysis),
- Build predictive models for clinical practice, genomics, epidemiology, etc.

Univariate Statistic, Multivariate Statistic, Machine Learning, Deep Learning → MODEL

Classical statistical techniques:

- **Univariate:** looks at a single variable (e.g. average days of hospitalization),
- **Multivariate:** examines relationships between multiple variables at the same time (e.g. age, comorbidity, therapy).

Usage → Data description, preliminary associations.



IL MODELLO MULTIVARIATO

L'equazione da stimare diventa (in notazione scalare):

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_k X_k + \varepsilon$$

dove le X_j sono le variabili indipendenti (o regressori) e i beta sono parametri (sconosciuti) oggetto di stima.

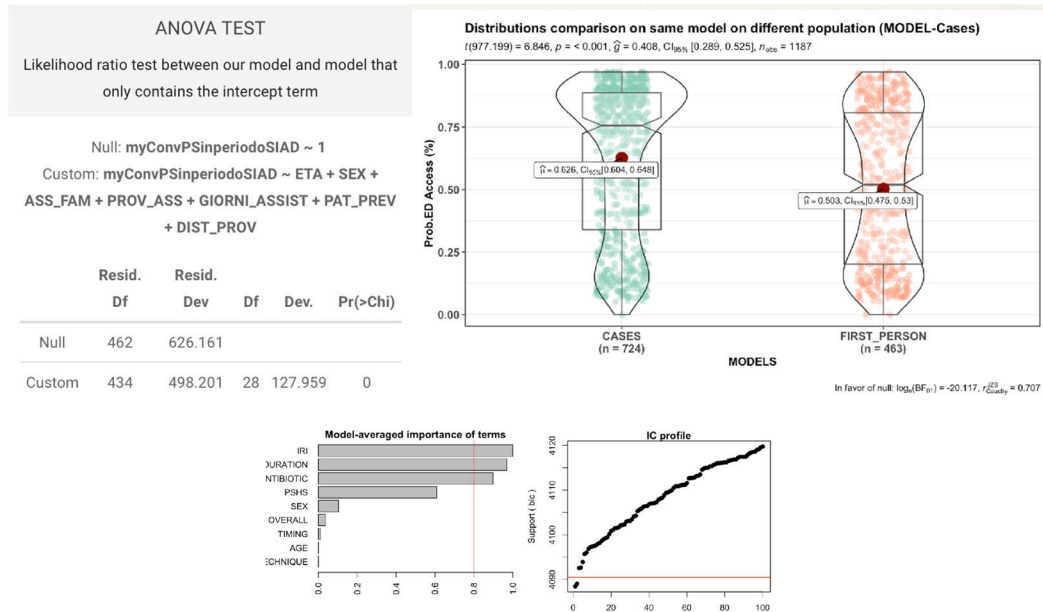
La logica OLS è la stessa

NB: qual è ora l'interpretazione dei beta?

Possono essere visti come derivate parziali: misurano cioè l'effetto sulla variabile dipendente di variazioni delle relative variabili indipendenti (ceteris paribus).

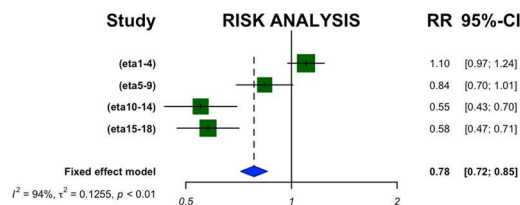
➤ GLM - Generalized Linear Models

- Linear Regression Extension: Allows to handle non-normal data, such as:
 - binary data (e.g. recovered/yes-no),
 - counts (e.g. number of infections),
 - proportions.
- Usage → Inference, prediction, evaluation of effects (e.g. impact of a surgical intervention on infections, as in your cited study).



➤ Meta Analysis

- Technique for combining the results of multiple independent studies,
- Provides a global estimate of the effect (e.g. risk, benefit),
- Usage → General validation, clinical guidelines.



➤ Fixed Effect Estimation

- Method used in longitudinal or panel data: controls for unobserved individual variables that remain constant over time,
- Usage → Trajectory studies, within-subject effects (e.g., emergency room visits in palliative patients over time).

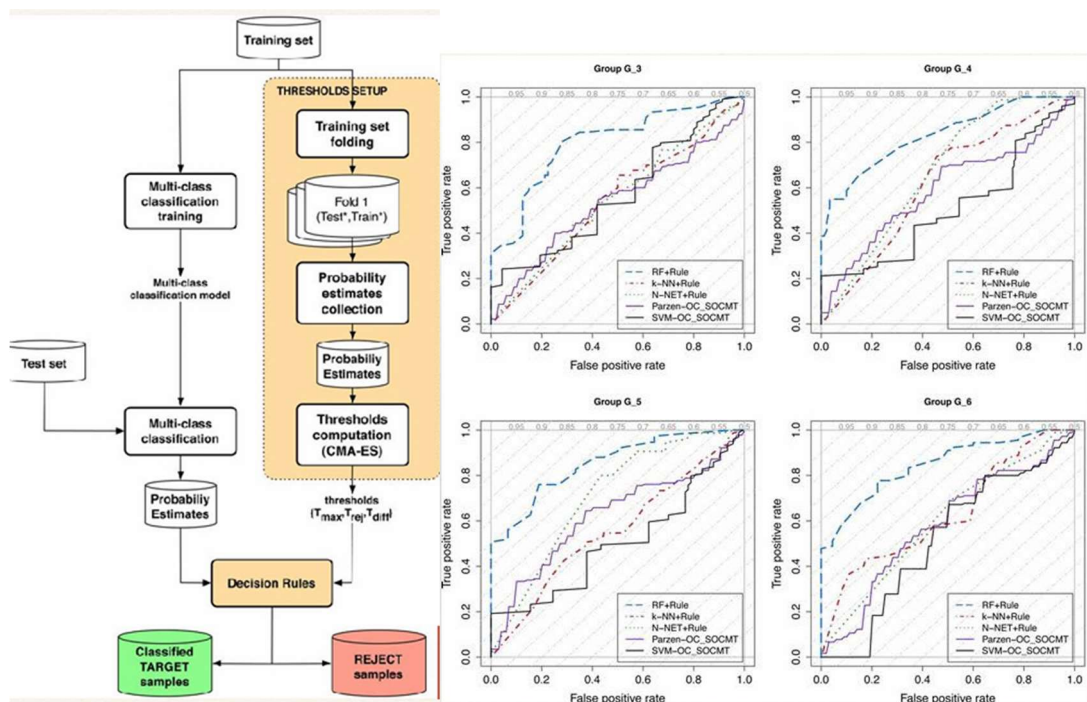
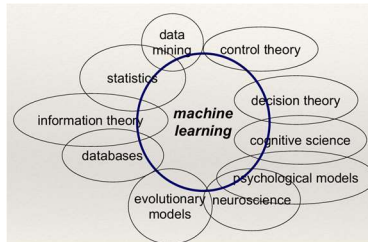
PSxSIAD_1 vs PROV_ASS - Models Comparison

Regressions with robust Standard Error

	Dependent variable: formulaOLS			
	OLS	panel linear		
	(OLS)	(Fixed)	(Random)	(Pooled)
		NA	NA	NA
PROV_ASS_PA01	-0.141 p = 0.549	0.225 p = 0.409	-0.026 p = 0.903	-0.141 p = 0.492
PROV_ASS_PA02_06_08_09	-0.064 p = 0.761	0.321* p = 0.098	0.057 p = 0.762	-0.064 p = 0.752
Constant	36.761* p = 0.069		26.197 p = 0.133	36.761** p = 0.034
Observations	70	70	70	70
R ²	0.013	0.043	0.011	0.013
Adjusted R ²	-0.017	-0.223	-0.019	-0.017
Residual Std. Error	7.830 (df = 67)			
F Statistic	0.426 (df = 2; 67)	1.201 (df = 2; 54)	0.738	0.426 (df = 2; 67)

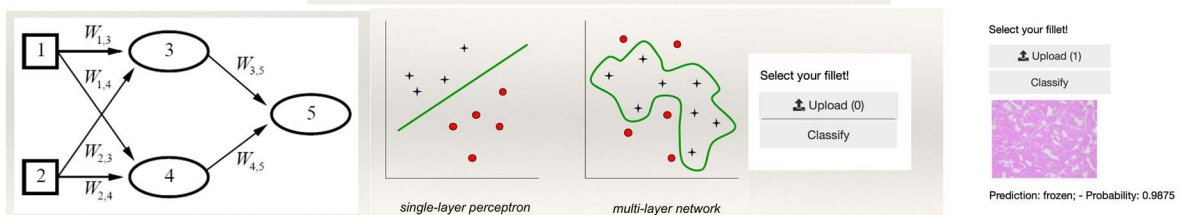
➤ Classification (ML):

- type of feedback
 - o supervised (labeled examples),
 - o unsupervised (unlabeled examples),
 - o reinforcement (reward),
- representation
 - o attribute-based (feature vector),
 - o relational (first-order logic),
- use of knowledge
 - o empirical (knowledge-free),
 - o analytical (knowledge-guided).



➤ Deep Learning:

$$a_5 = g(W_{3,5} \cdot a_3 + W_{4,5} \cdot a_4) \\ = g(W_{3,5} \cdot g(W_{1,3} \cdot a_1 + W_{2,3} \cdot a_2) + W_{4,5} \cdot g(W_{1,4} \cdot a_1 + W_{2,4} \cdot a_2))$$

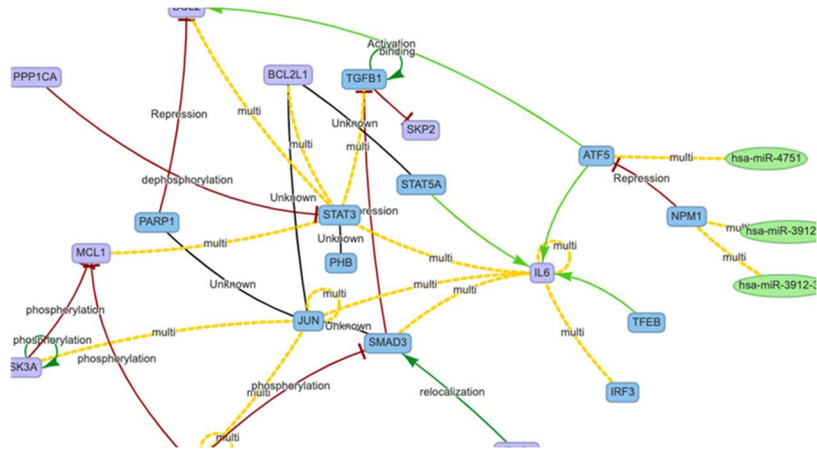


neural network = parameterized family of nonlinear functions

- types:
 - o Feed-forward (acyclic): single-layer perceptrons, multi-layer networks,
 - o recurrent (cyclic): Hopfield networks, Boltzmann machines.

➤ **Graph Theory:**

- advanced techniques,
- Usage → Analysis of biological networks (interactomes, pathways), COVID contacts, health networks.



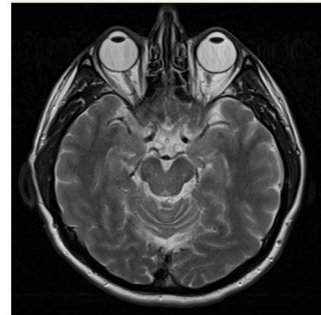
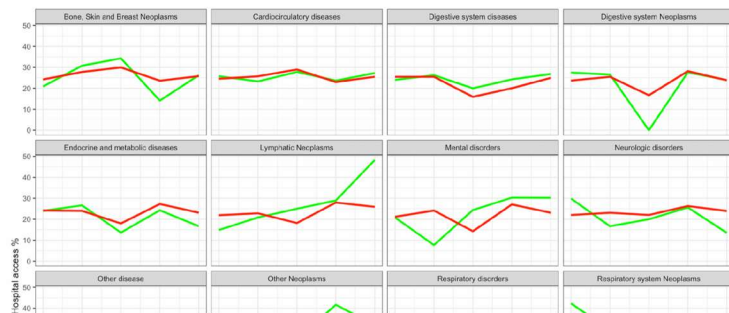
All these techniques converge on a common goal the predictive model.

➤ Models: Prediction

- Integrate different data sources and techniques,
- Generate accurate, reliable, clinically useful predictions,
- Personalize patient care (e.g. precision medicine),
- Support decisions (e.g. which patients are at risk of complications?).

PSxSIAD_1 vs PROV_ASS - PREDICTION

Green line: Raw Data; Red line: Prediction



disease?

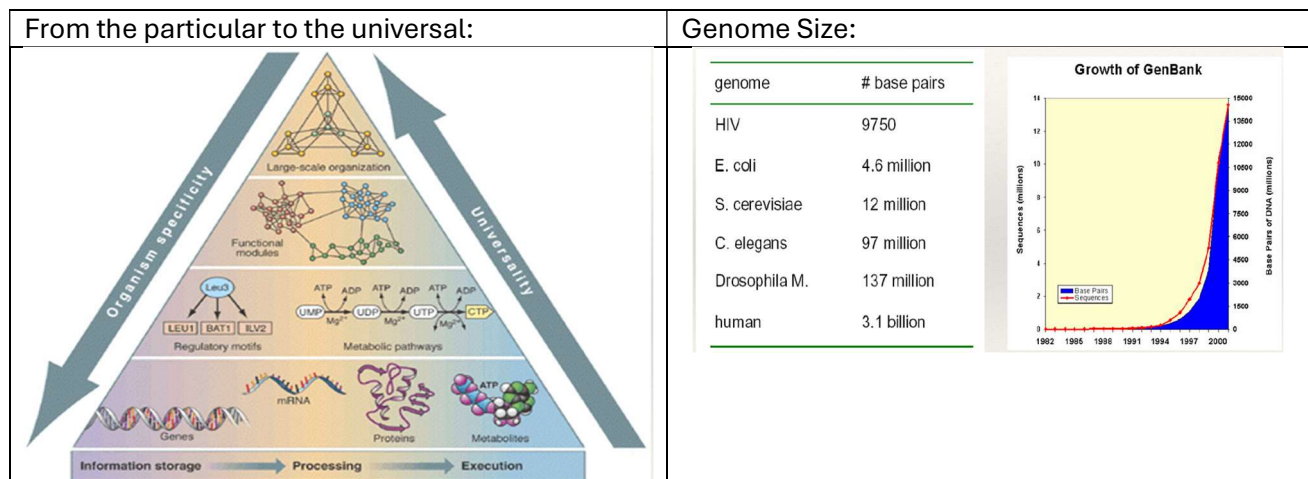
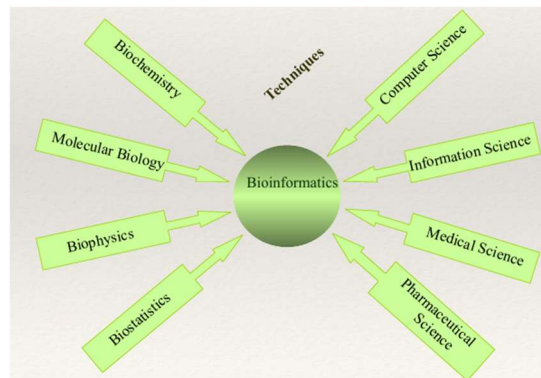
Biological network

AGENDA

- I. Basics on Bio,
- II. **Type of Network,**
- III. Pathways,
- IV. Databases,
- V. **Application:**
 - a. Network reconstruction,
 - b. Markov chain clustering,
 - c. Data integration,
 - d. Network based classification,
 - e. Network alignment,
- VI. PPI Networks Properties.

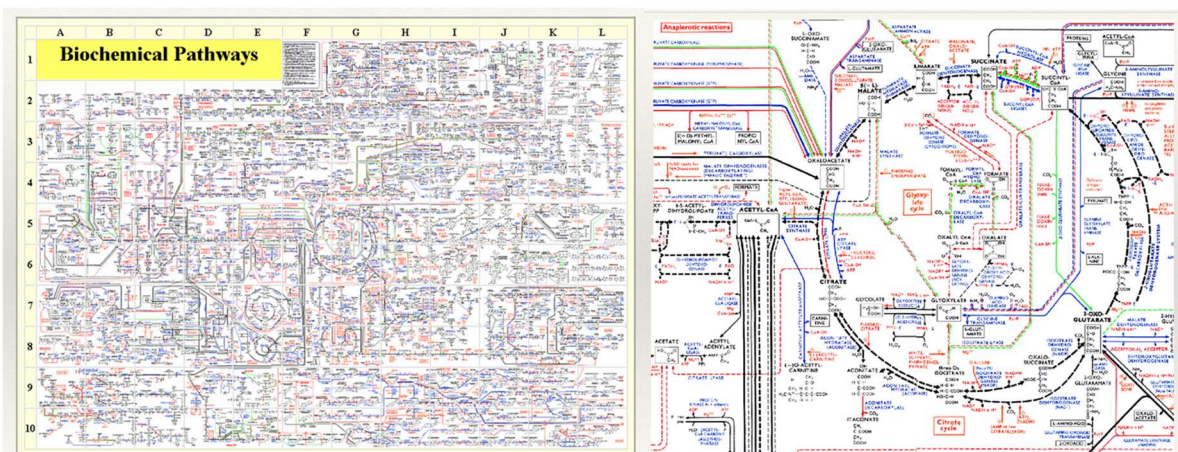
BASICS on Bio

Bioinformatics: interdisciplinary research area to manage and analyze biological data.



Networks are found in biological systems of varying scales:

1. Evolutionary tree of life,
2. Ecological networks,
3. Expression networks,
4. Regulatory networks - genetic control networks of organisms,
5. The protein interaction network in cells,
6. The metabolic network in cells ... more biological networks.



Why Study Networks?

- Complex systems cannot be described in a reductionist view,
- Understand their behaviour by understanding the topology of the corresponding network,
- Topological information is fundamental in constructing realistic models for the function of the network.

- Interaction Network,
- Pathway Network,
- Regulatory Network,
- Expression Network.

Biological Network Model

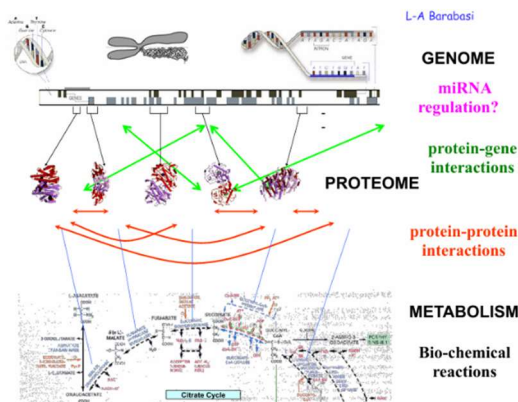
- **Network:** A linked list of interconnected nodes.
- **Node:** Protein, peptide, or non-protein biomolecules.
- **Edges:** Biological relationships, etc., interactions, regulations, reactions, transformations, activation, inhibitions.

A circle indicates a protein or a non-protein biomolecule. A symbol in between indicates the nature of molecule-molecule process (activation, inhibition, association, disassociation, etc.). It is usually represented by a **2-D diagram** with characteristic symbols linking the protein and non-protein entities.

Proteins in a cell

There are thousands of different active proteins in a cell acting as:

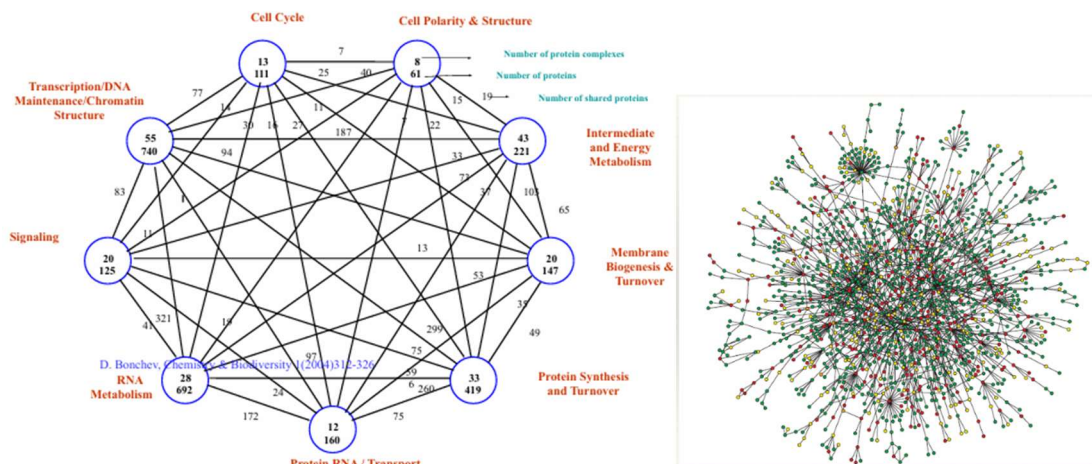
- enzymes, catalysors to chemical reactions of the metabolism,
- components of cellular machinery (e.g. ribosomes),
- regulators of gene expression,
- certain proteins play specific roles in special cellular compartments,
- others move from one compartment to another as “signals”.



Yeast

Yeast: 1400 proteins, 232 complexes, nine functional groups of complexes.

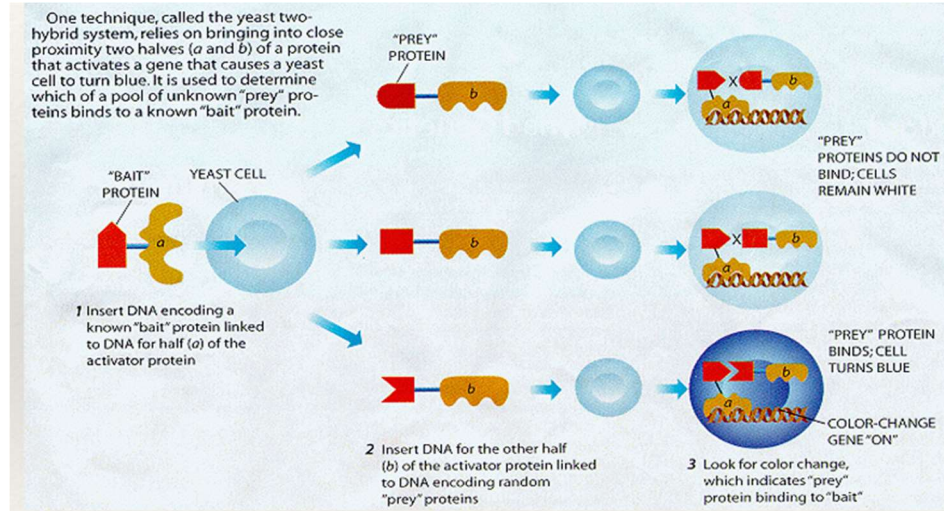
- ➔ Nodes: proteins,
- ➔ Links: physical interactions (binding).



Protein Interactions

Proteins perform a function as a complex rather than as a single protein. Knowing whether two proteins interact can help us discover unknown proteins' functions: if the function of one protein is known, the function of its binding partners are likely to be related- "guilt by association". Thus, having a good method for detecting interactions can allow us to use a small number of proteins with known function to characterize new proteins.

Finding proteins that interact:

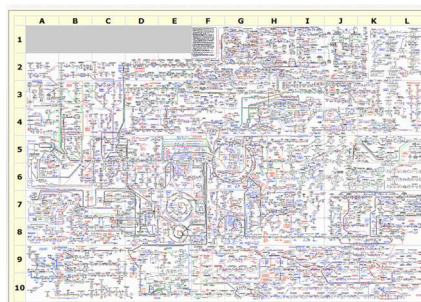


Problems

- Network Inference: Micro Array, Protein Chips, other high throughput assay methods,
- Function prediction:
 - o The function of 40-50% of the new proteins is unknown,
 - o Understanding biological function is important for:
 - Study of fundamental biological processes,
 - Drug design,
 - Genetic engineering,
- Essential Component Analysis,
- Functional module detection: Cluster analysis,
- **Topological Analysis:** Descriptive and Structural, Locality Analysis, Essential Component Analysis,
- **Dynamics Analysis:** Signal Flow Analysis, Metabolic Flux Analysis, Steady State, Response, Fluctuation Analysis,
- **Evolution Analysis.**

Biological Networks are very rich networks with very limited, noisy, and incomplete information. Discovering underlying principles is very challenging.

Boehringer Map:

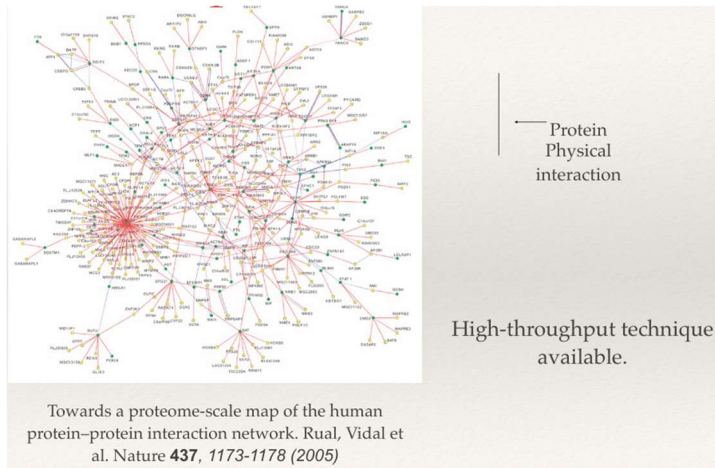


- ➔ A famous map of metabolic pathways, very detailed. It graphically shows the connections between the main metabolic reactions of a cell, it's done by hand.

Types of Networks

Interaction Network

- **Protein-protein interaction PPI:** This type of network describes which proteins interact with each other. Interactions can form protein complexes, signaling pathways, or functional modules. It helps to understand how proteins work together to perform biological functions.



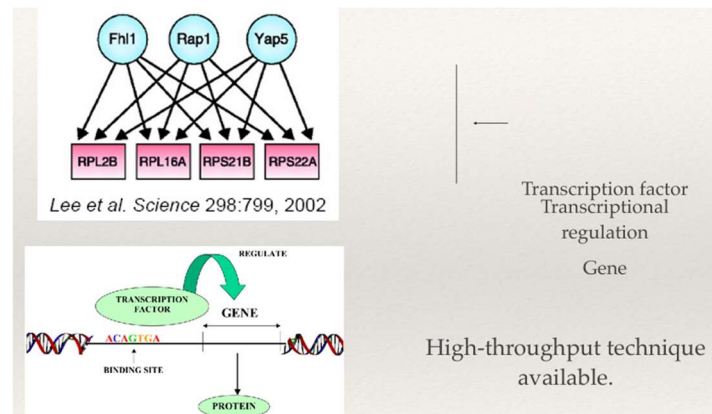
→ Few central points

- **Gene Regulatory Network - GRN:** It is a network that shows how genes regulate the expression of other genes. It models the control of cellular processes (e.g. cell cycle, embryonic development).

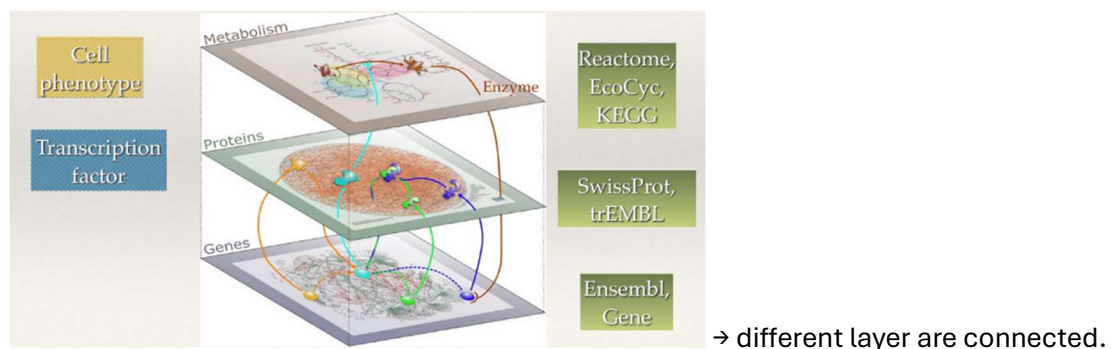
Example: a gene A produces a protein that activates gene B → A regulates B.

It includes:

- Regulatory genes (e.g. transcription factors),
- Target genes,
- Activating or inhibiting interactions.



General perspective: A biological function is a result of many interacting molecules and cannot be attributed to just a single molecule.



Pathway Networks

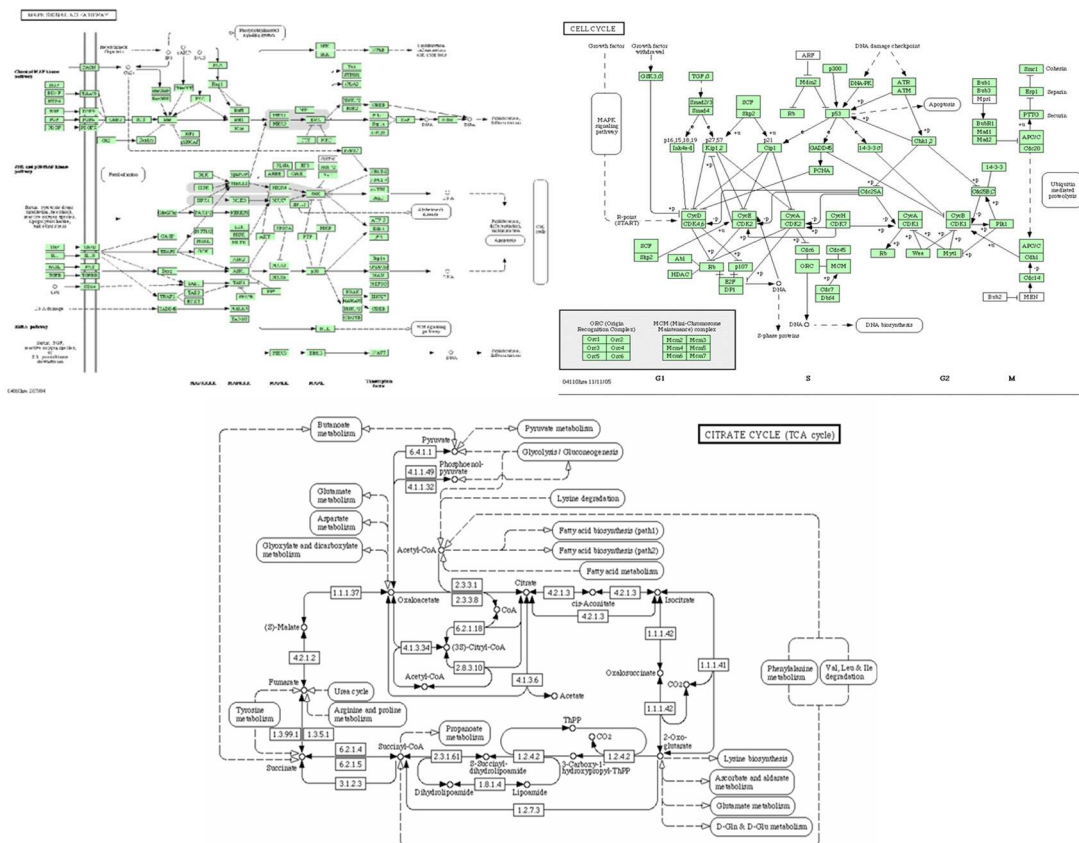
(metabolic) Pathway

- Wikipedia: series of chemical reactions occurring within a cell,
- Reactome: any related set of events,
- KEGG: set of reactions centered on the synthesis and/or degradation of one or more related substrates,
- BioCyc: single, evolutionary conserved, most biological process of which reactions are active at the same time.

A **Pathway** can be defined as a modular unit of interacting molecules to fulfil a cellular function. It also defines as: an ordered sequence of biological reactions that occur within a cell that lead to a certain biological outcome

- **Signaling Pathway Networks:**
 - o In biology a **signal** or biopotential is an electric quantity (voltage or current or field strength), caused by chemical reactions of charged ions. Signals refer to any process by which a cell converts one kind of signal or stimulus into another. Another use of the term lies in describing the transfer of information between and within cells, as in signal transduction. It is also defines as a networks model cell signaling processes, that is, how cells receive and respond to external or internal signals.
- **Metabolic Pathway Networks:**
 - o A series of chemical reactions occurring within a cell, catalyzed by enzymes, resulting in either the formation of a metabolic product to be used or stored by the cell, or the initiation of another metabolic pathway so a metabolic pathway network is a network of interconnected chemical reactions that occur within a cell.

A Pathway Example:

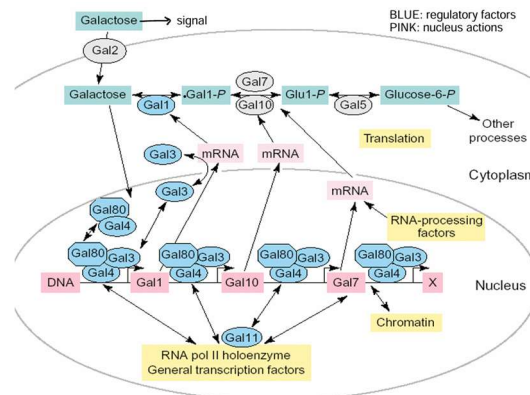


➔ It's not more represented as cell, there are numbers. Selecting some of the element you can obtain more information about the specific selection.

Regulatory Network

Regulatory Network is a collection of DNA segments (genes) in a cell which interact with each other and with other substances in the cell, thereby governing the rates at which genes in the network are transcribed into mRNA.

Note: the rate is unknown, it can be very fast or very slow.

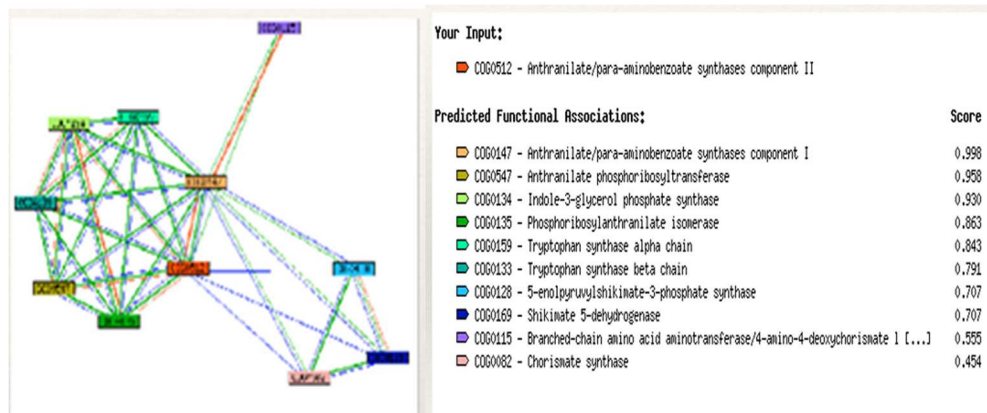


➔ Genomic network:

- No chemical elements,
- Direct connections,
- Cell representation.

Expression Network

A network representation of genomic data. Inferred from genomic data, i.e. microarray. It is construct using genomic data like microarray and RNA-seq.



- It's not show causal connection but probabilistic connections,
- The connections are based on the expression correlation, if two genes are expressed similarly under many conditions they become connected,
- Less organization → less sense.

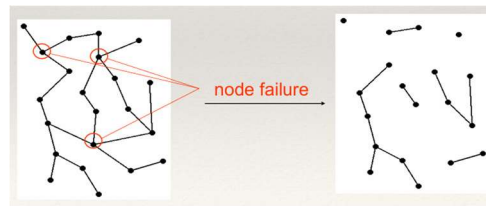
Note: gene ontology, a way to organize knowledge in a hierarchical manner.

Biological Network Property

- **Small World:** A small average path length, mean shortest node-to-node path.
- **Robustness:** Resilient and have strong resistance to failure on random attacks and vulnerable to targeted attacks.
- **Hierarchical Modularity:** A large clustering coefficient. How many of a node's neighbors are connected to each other.

Attack Tolerance

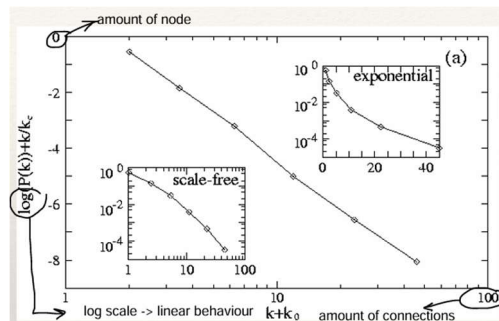
Complex systems maintain their basic functions even under errors and failures (cell → mutations; Internet → router breakdowns).



Protein Interaction Network

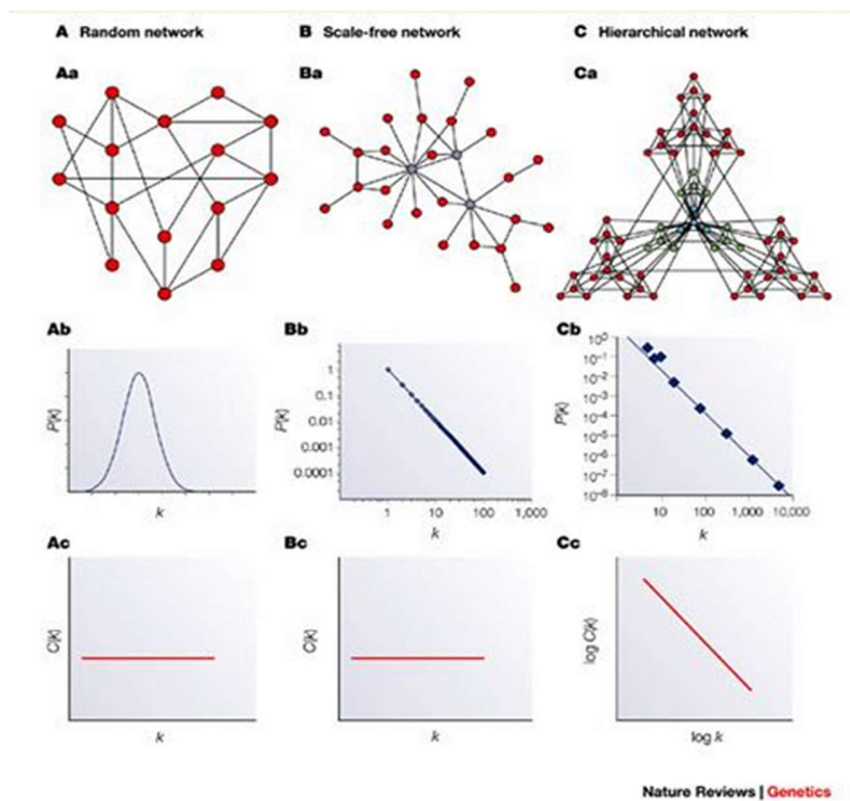
The yeast protein interaction network seems to reveal some basic graph theoretic properties:

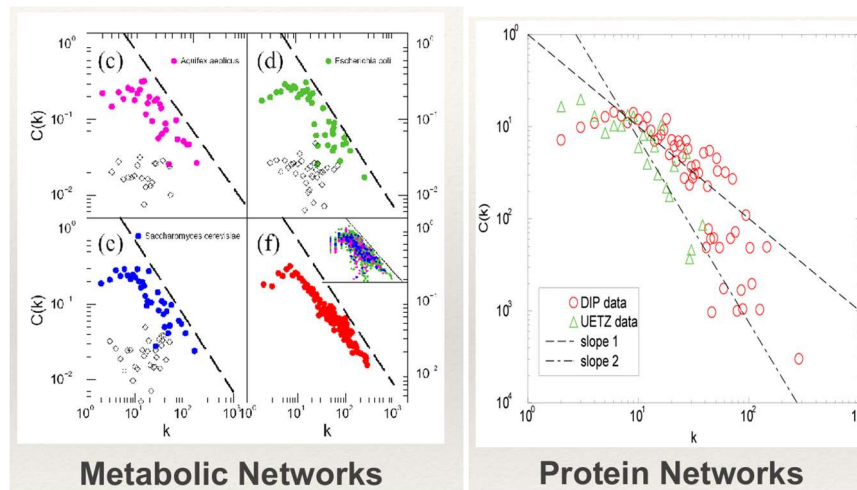
- The frequency of proteins having interactions with exactly k other proteins follows a power law,
- The network exhibits the small world phenomena: can reach any node within small number of hops, usually 4 or 5 hops, more hubs we have less jump we need to go through the graph,
- Robustness: Resilient and have strong resistance to failure on random attacks and vulnerable to targeted attacks.



Note: protein-to-protein interactions there isn't direction (is bidirectional), gene interaction has directionality

Hierarchical Modularity





Implications From Observations

- Biological complexity: # states $\sim 2^{\text{#of genes}}$, due to each gene can be ON-OFF,
- Protein hubs critical for cells, 45%,
 - o Hub: proteins that interact with many others,
 - o 45%: this is the percentage of essential proteins for living cells,
- Infections will target highly connected nodes, it's better to have problem at downstream level as low as possible,
- Cascading node failures could cause a critical problem.
- Development of drug and treatment with novel strategies like targeting effective nodes is indispensable.

Network Databases

Protein Interaction Databases

- Swiss-Prot (non-redundant database): >> 124,464 entries
- TrEMBL (translations of EMBL nucleotide sequences not yet integrated into Swiss-Prot): million entries,
- This number keeps rapidly growing mainly due to large scale sequencing projects.

Species-specific:

- FlyNets - Gene networks in the fruit fly,
- MIPS - Yeast Genome Database,
- RegulonDB - A DataBase On Transcriptional Regulation in E. Coli,
- SoyBase,
- PIMdb - Drosophila Protein Interaction Map database,
- CSNDB - Cell Signaling Networks Database.

Function-specific:

- Biocatalysis/Biodegradation Database,
- BRITE - Biomolecular Relations in Information Transmission and Expression,
- COPE - Cytokines Online Pathfinder Encyclopaedia,
- Dynamic Signaling Maps,
- EMP - The Enzymology Database,
- FIMM - A Database of Functional Molecular Immunology,
- CSNDB - Cell Signaling Networks Database.

Interaction type-specific:

- DIP - Database of Interacting Proteins,
- DIPInteract - DNA-protein interactions,
- Inter-Chain Beta-Sheets (ICBS) - A database of protein-protein interactions mediated by interchain beta-sheet formation,
- Interact - A Protein-Protein Interaction database,
- GeneNet (Gene networks).

General:

- BIND - Biomolecular Interaction Network Database,
- BindingDB - The Binding Database,

- MINT - a database of Molecular INTERactions,
- PATIKA - Pathway Analysis Tool for Integration and Knowledge Acquisition,
- PFBP - Protein Function and Biochemical Pathways Project,
- PIM (Protein Interaction Map).

Pathway Databases

- KEGG (Kyoto Encyclopedia of Genes and Genomes)
 - o <http://www.genome.ad.jp/kegg/>
 - o Institute for Chemical Research, Kyoto University
- PathDB
 - o <http://www.ncgr.org/pathdb/index.html>
 - o National Center for Genomic Resources
- SPAD: Signaling PATHway Database
 - o Graduate School of Genetic Resources Technology. Kyushu University.
- Cytokine Signaling Pathway DB.
 - o Dept. of Biochemistry. Kumamoto Univ.
- EcoCyc and MetaCyc
 - o Stanford Research Institute
- BIND (Biomolecular Interaction Network Database)
 - a. UBC, Univ. of Toronto.

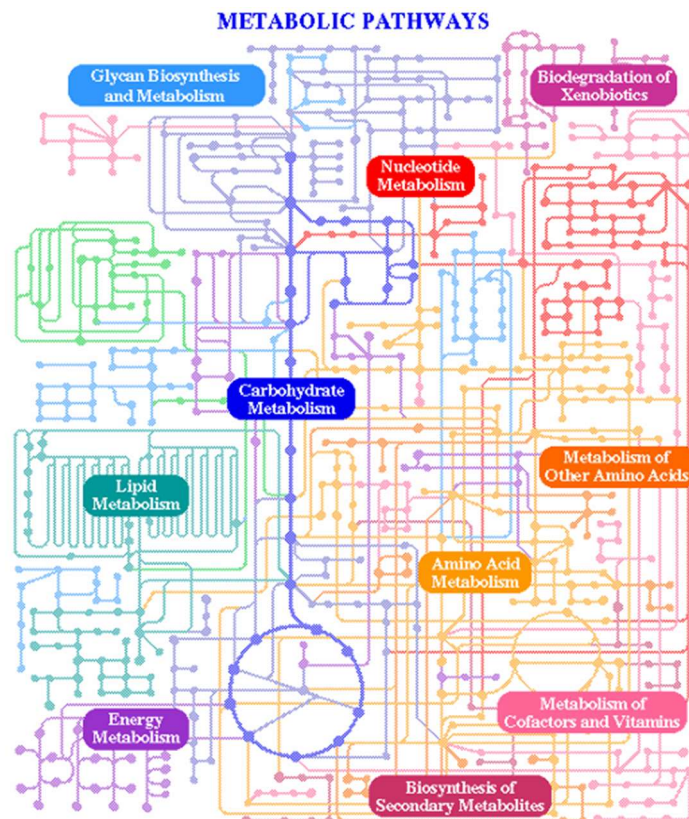
KEGG

Pathway Database: Computerize current knowledge of molecular and cellular biology in terms of the pathway of interacting molecules or genes.

Genes Database: Maintain gene catalogues of all sequenced organisms and link each gene product to a pathway component.

Ligand Database: Organize a database of all chemical compounds in living cells and link each compound to a pathway component.

Pathway Tools: Develop new bioinformatics technologies for functional genomics, such as pathway comparison, pathway reconstruction, and pathway design.



Applications

- Network reconstruction,
- Markov chain clustering,
- Data integration,
- Network based classification,
- Network alignment.

Network Reconstruction

Networks consist of interactions. Interactions can be based on many data sources (different source in different time):

- **Direct:** from a experiment
 - o protein-DNA interaction (ChIP),
 - o protein-protein interaction (yeast two-hybrid),
 - o protein complexes (mass spectrometry),
- **Indirect:** often models are needed (=network reconstruction):
 - o gene co-expression (microarray),
 - o co-citation (text mining), the best single source of information,
 - network: gene in papers, if genes appear in the same paper create an edge, for each (co-)appearance increase the weight of the edge.

Microarray expression data:

Reconstruct a model of the gene network using expression data.

Pros of using microarrays:

- Less complex experiments,
- Global overview of the whole genome in a single experiment. e.g. one mass spectrometry experiment identifies only about 2000 proteins.

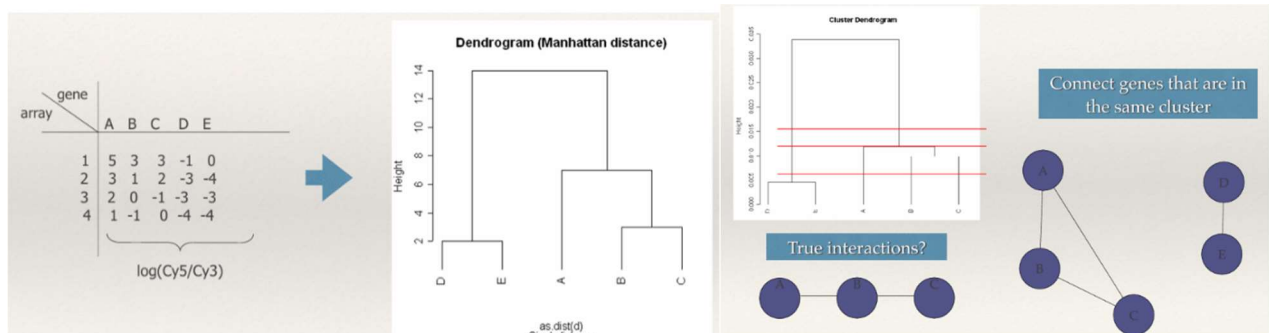
Assumptions:

- mRNA abundance levels ~ protein concentrations,
 - o If a gene is highly expressed → it is assumed that its protein is present in large quantities,
- co-expression ~ co-regulation,
 - o If two genes are always active together → they are probably regulated by the same mechanism,
- co-expressed genes ~ interacting products (proteins) ~ transcription factor – gene,
 - o If genes are co-expressed → their products (proteins) could interact.

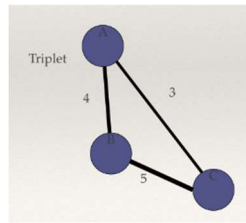


Hierarchical clustering for network reconstruction:

Note: they are techniques for microarray reconstruction.



ARACNE - Algorithm for the Reconstruction of Accurate Cellular Networks:

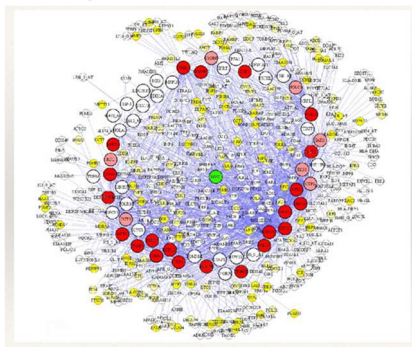


Method for reconstructing biological network using **information theoretic method** (used to measure how much 2 genes are dependent to each other) **to reduce false positives** which are predicted through indirect interactions. Identifies statistically significant gene-gene coregulation by mutual information. It then eliminates indirect relationships in which two genes are coregulated through one or more intermediates. Examine each triplet Information theory says that: The two genes that do NOT directly interact have the lowest Mutual Information. → Remove the edge with the lowest Mutual Information.

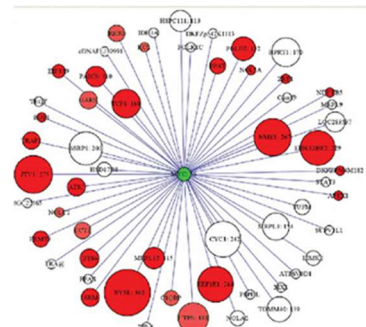
Ex: $A \rightarrow B, B \rightarrow C \Rightarrow A \rightarrow C$ can be a false link (remove it)

Applied to B lymphocytes expression data: MYC (proto-oncogene) subnetwork (2063 genes), 29 of the 56 (51.8%) predicted first neighbours biochemically validated as targets of the MYC transcription factor. New candidate targets were identified, 12 experimentally validated. 11 proved to be true targets. The candidate targets that have not been validated are possibly also correct.

B lymphocytes expression data – result:



→

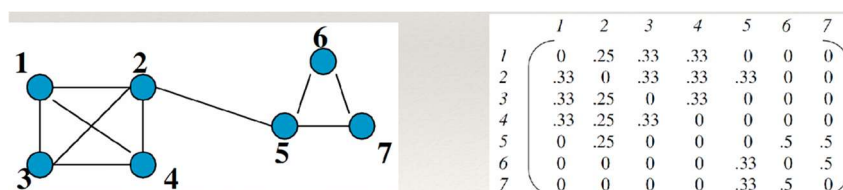


Markov Chain Clustering (MCL)

Graph clustering algorithm, many links within cluster and fewer between clusters by doing random walks upon the graph, it may be possible to discover where the flow tends to gather, and therefore, where clusters are. Random Walks on a graph are calculated using “Markov Chains”. The MCL algorithm finds cluster structure in graph by a mathematical bootstrapping procedure. Simulates random walks within a graph by alternations of two operators **expansion** (simulates longer random walks in the graph) and **inflation** (applies a nonlinear enhancement followed by a normalization).

- ➔ Higher step transition probabilities are obtained,
- ➔ large transition probabilities are promoted and small demoted.

In one time step, a random walker at node 1 has a 33% chance of going to node 2, 3, & 4, and 0% chance to nodes 5, 6, or 7. From node 2, 25% chance for 1, 3, 4, 5 and 0% for 6 and 7. Creating a transition matrix gives:



DATA INTEGRATION

Notes of caution: a lot of false positives and false negatives.

Artificial circumstances - uncertainties:

- Proteins may not be in the same cellular location.
- They may not be present at the same time.
- The experiments did not capture the specific circumstance.

The detected interaction might be a coincidence as protein binding is a stochastic process. The binding of a transcription factor does not necessarily mean that the gene will be transcribed.

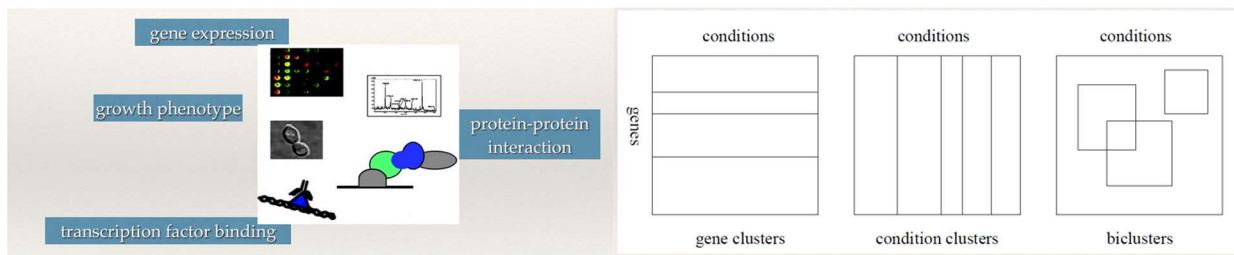
Integrating experimental data

Instead of using only one type of data use several types.

This will:

- give more supporting evidence that a protein performs a certain function,
- reduce the number of false positive and false negative interactions,
- give a more complete picture of the interactions between different elements involved in a certain biological process.

Statistical-Algorithmic Method for Bicluster Analysis

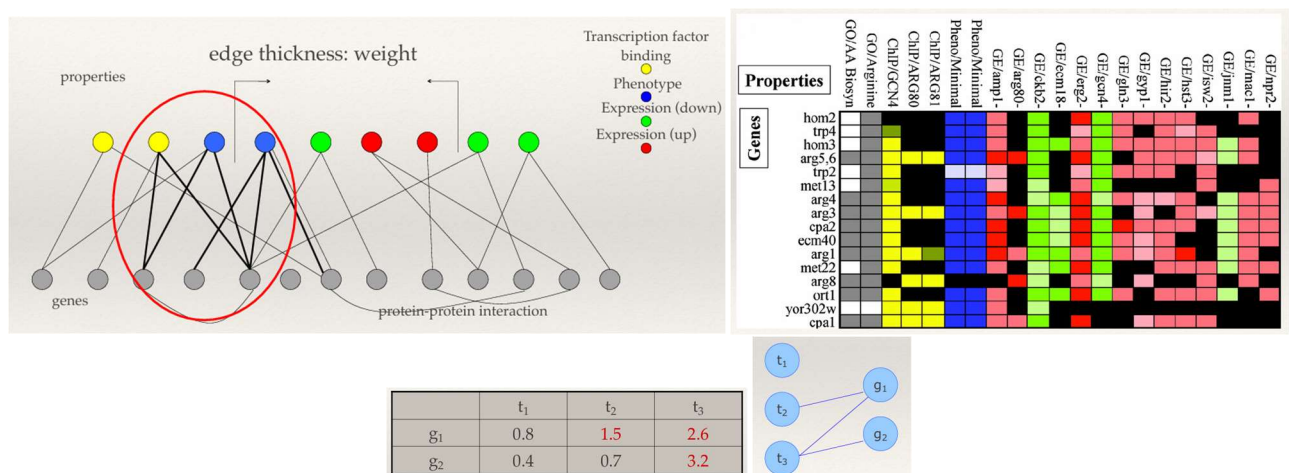


A **bicluster** is defined as a submatrix spanned by a set of genes and a set of samples.

Model: The whole dataset forms a **bipartite graph** $G = (U, V, E)$:

- U is the set of conditions,
- V is the set of genes,
- $(u, v) \in E$ if v responds in condition u (i.e., the expression level of v changes significantly in u)
- A **bicluster**: a subgraph of the bipartite graph.

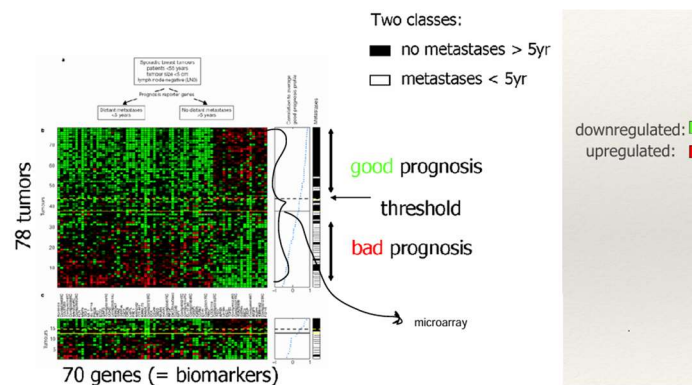
Example: 1000 genes * 100 conditions, if 20 genes have the same behaviour in 5 conditions, this is a bicluster.



→ YOR302W is included because of TF binding & phenotype. Experimental evidence: YOR302W regulates the translation of the CPA1 mRNA.

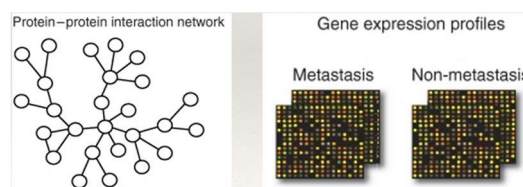
NETWORK-BASED CLASSIFICATION

Classification using gene expression profiles: the goal is to improve classification not only using gene expression data but also using protein-protein interaction (PPI) network.



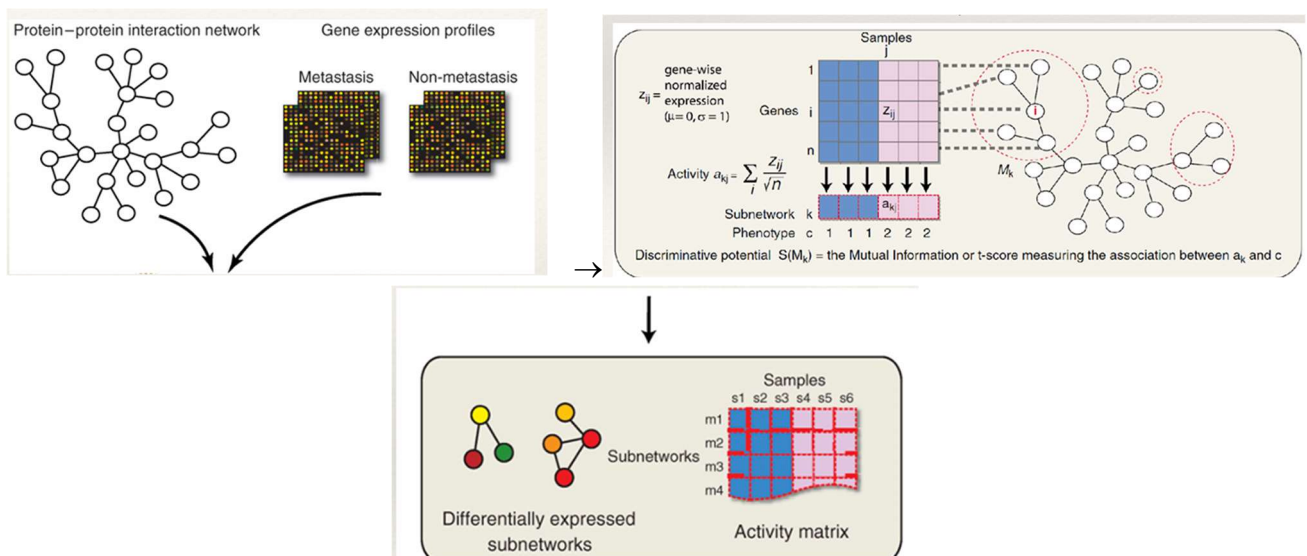
Chuang et al. and Benso et al proposed to combine:

- expression data also using a **protein-protein interaction network** - Chuang,
- **Gene expression Graph** - Benso.



Method:

- Overlay the expression value of each gene on its corresponding protein,
- Search for subnetworks for which the score is maximal (ex: most significant differences between healthy patients and weak patients),
- Select significant networks based on null distributions estimated from *permuted* subnetworks,
- Use (the subnetwork) for *classification* of new samples/patients.



Advantages:

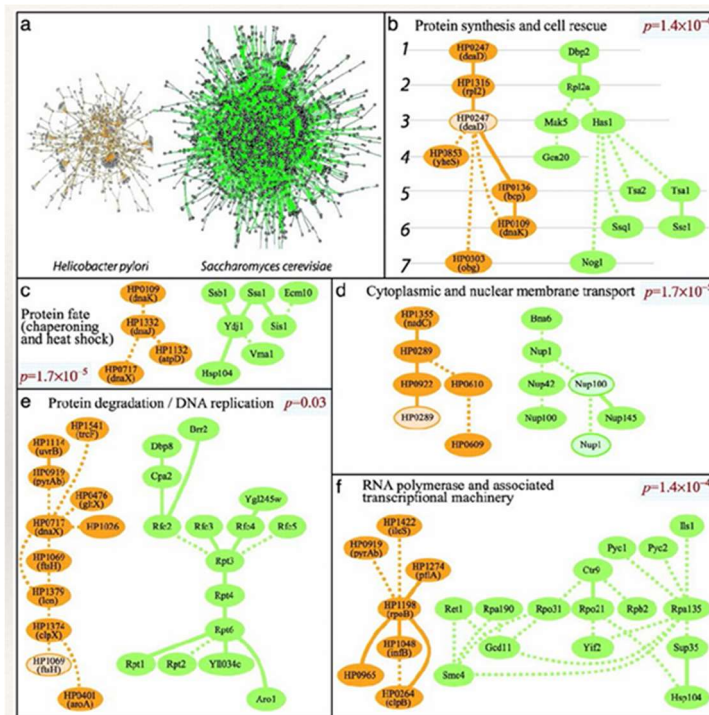
- Genes not differentially expressed can be included if connected to many genes that are differentially expressed. This included known breast cancer mutations, phenotypic change indicative of metastasis might not be measured at the level of gene expression,
- The subnetworks give small scale models of molecular mechanisms underlying metastasis,
- The subnetworks are more robust biomarkers than single genes.

Network Alignment

Alignment: “The global comparison of two networks to identify regions of similarity and dissimilarity.”

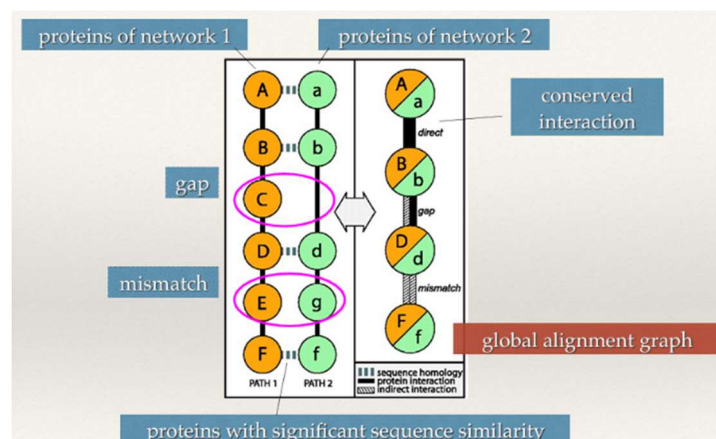
A network alignment can be used for:

- Function prediction, predicting the function of proteins in one species based on another,
- Drug development: identifying the interactions that are common between pathogen and host and develops new drugs,
- Reduce the number of false positive interactions: Unlikely that e.g. incorrect protein-protein interaction will be found in 5 different PPI networks of different experiments or organisms if an interaction is confirmed in multiple networks → more reliable,
- Evolution of molecular process.



PathBLAST

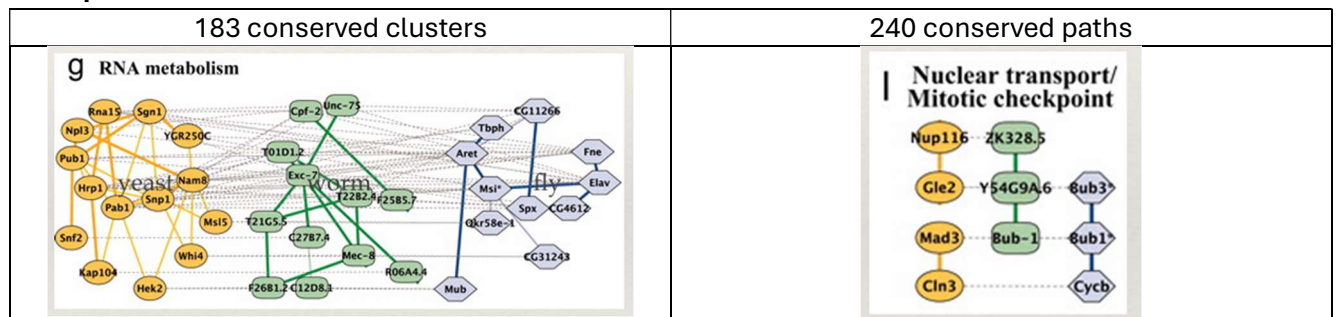
1. Use BLAST to determine the sequence similarity between the proteins,
2. Make a global alignment graph. Align proteins that share sequence similarity,
3. Search for high-scoring alignments:
 - a. Scoring function is a combination of the probabilities that the:
 - i. proteins are true homologs (look like the same),
 - ii. interaction is a true positive (true interaction).
 - b. Use dynamic programming to find high-scoring paths (best path) of a fixed length.



NetworkBLAST

Extension of PathBLAST using a similar scoring function and global alignment graph. More than two networks, nodes now contain a protein from each network. Each protein has to be similar to at least one other protein. Identify high-scoring subgraphs and paths (not only high-scoring path). Start by finding high-scoring paths of length 4. Extend by adding and removing nodes to improve the score.

Examples of results



Network alignment VS Sequence alignment

- Higher level overview,
- Predicting of functions using alignment:
 - o NetworkBLAST: 58% – 63% correct
 - Identify conserved interactions as well as sequences across species.
 - Integrate a wide variety of biological data,
 - NetworkBLAST reaches its practical limit at 4 or 5 networks
 - o Sequence-based method; 37% – 53% correct,
 - BLAST can be done against a large number of organisms.

Difficulties network alignment:

- As opposed to sequences, networks are not linear structures.
- No efficient solutions for aligning networks, optimal network alignment is an NP-complex problem and the complexity increase $\propto e^{\#networks}$.

Recap

Network reconstruction: Less false positives without losing many true positives. However, you cannot determine the direction of the interaction and from a loop of three genes one edge will always be removed.

Data integration gives a more complete picture of a certain biological process.

Sub-network markers are more reproducible than individual marker genes and that they achieve higher accuracy in the classification.

Pair-wise and multiple alignment of networks.

PPI Networks Properties

Biological Network

Definition:

- Directed or undirected graph representation,
- **Biological molecules** as nodes and **biochemical reactions** or biophysical interactions as edges.

Examples: Metabolic networks, Signal transduction networks, Gene regulatory networks, Protein interaction networks.

Importance:

- Provide a global view of cellular organizations and biological processes,
- Applicable to systematic approaches for knowledge discovery.

Protein-Protein Interaction (PPI)

Biological Meaning of PPI:

- Proteins interact with each other for stability and functionality,
- Most cellular functions are performed in a protein complex level,
- Interaction evidence is interpreted as functional coherence / consistency.

Determination of PPIs:

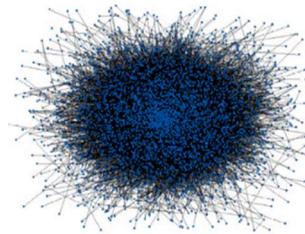
- Experimental methods,
- Computational methods.

Problem of PPI data:

- Current PPI databases include a large amount of false positives / false negatives,
- Unreliability.

Representation of Protein Interaction Networks:

- Undirected, un-weighted graph $G(V, E)$,
- a set of nodes V as proteins and a set of edges E as interactions.



Problem of Protein Interaction Networks:

- Large scale,
- Complex connectivity.

Structural Properties

Small-world Phenomenon (Watts & Strogatz):

- Appearance of networks in the middle of regular and random networks,
- **Higher average** clustering coefficient than expected by random chance,
- Significantly **small average shortest** path length.

Scale-free Distribution (Barabasi & Albert):

- Network growth by preferential attachment,
- **Power law degree distribution** implies robustness but vulnerability to targeted attacks,
 - o a few high degree nodes (few hubs with many connections),
 - o many low degree nodes (many nodes with few connections),
- Clustering coefficient distribution independent to degree.

Protein Interaction Database	DIP	MIPS
density	0.0015	0.0015
average clustering coefficient	0.2283	0.2878
average shortest path length	4.14	4.43
degree distribution (γ)	1.77	1.64

➔ high modularity, hub existence.

Network Weighting Schemes

Network weighting are used to improve the reliability of PPI networks but there are other motivations:

- Unreliable protein interaction networks,
- Transforming un-weighted graph to weighted graph,
- interaction reliability (or intensity) -> edge weight.

Unsupervised Approaches:

- Using network connectivity, e.g., common neighbors, → based on topology only,
- Problem: unreliable weights,

Supervised Approaches:

- Using other resources verifying interactions, e.g., gene sequence, gene expression,
- Integrating Gene Ontology data, the most comprehensive, well-curated.

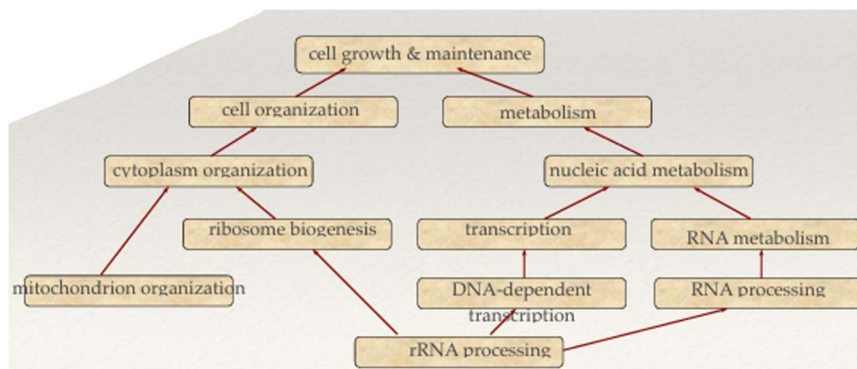
GENE ONTOLOGY - GO

Structure:

- Terms (Concepts): well-defined biological description,
- Relationships: “is-a” / “part-of” (general-to-specific) between terms.

Annotation:

- If a protein is annotated on a term, then it is also annotated on the terms on the paths towards root, this means that if two proteins have similar annotations → they are more likely to interact.



➔ Hierarchical structure, children have the same (and more) properties of the parents (father).

Semantic Similarity

Reliability of Interacting Proteins:

- Average (or Maximum) semantic similarity of pair-wise terms,
- Including the interacting proteins in annotations.

Structure-based Approaches:

- Path length or Common parent terms,
- Problem: all edges should represent the uniform specificity.

Information Content-based Approaches:

- Information content of a term T is defined as $-\log(P(T))$,
- $\text{sim}_{xy} = -\log(P_i(x,y))$ where $P_i(x,y)$ is the proportion of the annotations of the term including x and y
- Normalized $\text{sim}_{xy} = \frac{2 \times \log P_i(x,y)}{\log P_i(x) + \log P_i(y)}$

Functional Module Identification

Functional module is a set of molecules that participate in the same biological processes or functions, but also a sub-network with dense intra-connections and sparse interconnection.

To identify a functional module you can use Graph clustering problem.

Clustering Approaches:

- *Density-based methods*, e.g., maximum clique, quasi clique, clique percolation,
- *Partition-based methods*, e.g., restricted neighborhood search, Markov clustering,
- *Hierarchical methods*:
 - Bottom-up approaches, e.g., distance-based, common neighbors,
 - Top-down approaches, e.g., minimum cut, betweenness cut.

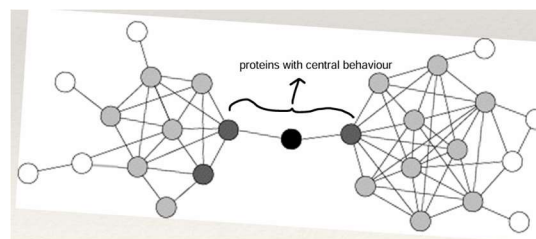
Hub Protein Identification

Hub Protein:

- Centrally located node in the modular structure of a protein interaction network (structural hub),
- Functionally essential protein, essential for living cells' proteins.

Previous Centrality Measurements:

- Closeness centrality, minimum average distance from other nodes,
- Betweenness centrality, how much a node is on shortest paths,
- Bridging centrality, measures the importance of connections between different modules.



→ if you cut out the darker nodes you can isolate the two network

Structural Hubs:

Hub	Degree	Functions
YJL081c	9	DNA conformation modification, transcription activation/repression, protein modification by (de)acetylation, organization of chromosome structure
YKL018w	14	transcription termination, post-translational modification of amino acid, organization of chromosome structure
YOL133w	8	mitotic cell cycle, transcription activation/repression, protein modification by (de)ubiquitination, proteasomal degradation, ATP binding
YHR061c	12	G-protein mediated signal transduction, pheromone response, directional cell growth, biogenesis of cytoskeleton, budding
YMR047c	41	tRNA processing, protein targeting, structural protein binding, protein transport, RNA transport, biogenesis of nuclear membrane
YEL037c	23	DNA repair, cell cycle checkpoints, enzyme inhibitor, DNA damage response, biogenesis of cell wall
YER133w	34	phosphate metabolism, regulation of C-compound, cell cycle checkpoints, meiosis, transcription termination, rRNA processing, translational control, protein modification by (de)phosphorylation, heat shock response, temperature perception, budding
YOR122c	5	protein binding, osmotic stress response, biogenesis of cytoskeleton, budding
YLR229c	16	GTP binding, exocytosis, G-protein mediated signal transduction, pheromone response, directional cell growth, biogenesis of cytoskeleton, budding
YBR160w	89	phosphate metabolism, DNA synthesis/replication, mitotic cell cycle, meiosis, protein modification by (de)phosphorylation, budding

Top 10 Structural Hubs in the Yeast Protein Interaction Network
 Not related to their degree
 Each one has several different functions

Lethality

Biological Essentiality:

- Evaluated by comparing with lethal proteins,
- Lethality can be determined by protein knock-out experiments.

Conclusion

Problems: complex and unreliable connectivity in protein interaction networks,

Contributions:

- Reliable network generation by edge weighting,

- Hidden knowledge discovery, e.g., patterns or taxonomy,
- Collaboration with existing computational techniques.

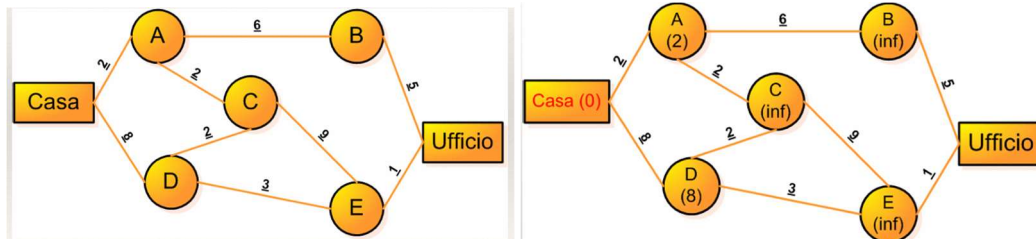
Goals:

- Integration with multiple data sources, as many source as you can,
- Comparative analysis across organisms.

The Dijkstra's algorithm

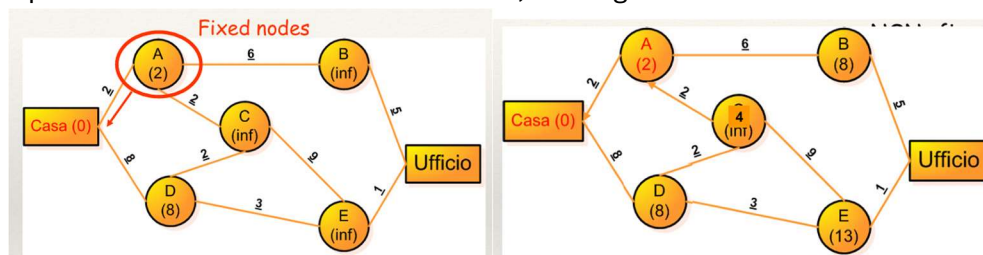
How to compute the shortest path between home and work? The exhaustive search can be too much time-consuming.

Initialization: Fix the distance between “Casa” and “Casa” equal to 0. Compute the distance between “Casa” and its neighbours. Set the distance between “Casa” and its NON-neighbours equal to ∞ .

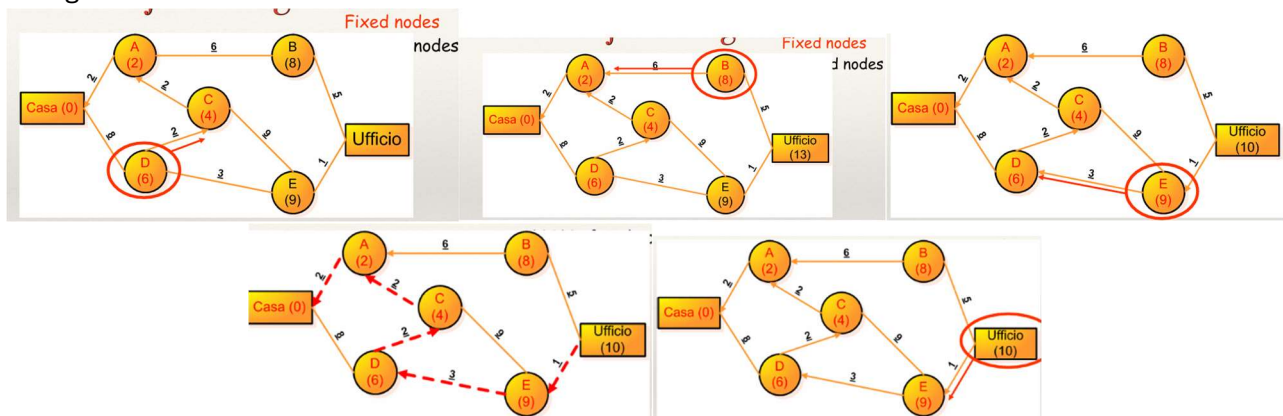


Iteration (1): Search the node with the minimum distance among the NON fixed nodes and Fix its distance, memorizing the incoming direction.

Iteration (2): Update the distance of NON-fixed nodes, starting from the fixed distances.



Iteration: Fix the NON-fixed nodes with minimum distance. Update the distance of NON-fixed nodes, starting from the fixed distances.



Conclusion: The label of each node represents the min. distance from the start node. The minimal path can be reconstructed with a back-tracing procedure.

Software – Cytoscape: a software used to analyse biological network.

